Fractals: an Understanding via Topological, Box, Similarity and Hausdorff Dimensions

by

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Abstract

This article undertakes a rigorous examination of four fundamental notions of dimension – topological, box-counting, similarity and Hausdorff dimensions, and their application to both classical and fractal sets. After introducing the necessary analytical and topological preliminaries, we compute and compare the dimensions of standard sets such as the interval $[0,1]$, the unit box $[0,1]^2$ and the unit circle S^1, as well as canonical fractals including the α-Cantor set C^α, the Sierpiński gasket S_3, the Sierpiński carpet S_4 and the Koch snowflake K. These computations highlight key distinctions among dimension concepts and illustrate how non-integer values naturally arise in the study of fractals. The results affirm the necessity of other notions of dimension for capturing the geometric complexity inherent in such sets.

Acknowledgements

I am truly grateful to my mentor, Dr. Siva Athreya, for his constant support and guidance throughout this project. His guidance, patience, and thoughtful feedback helped me understand concepts better and stay motivated. I have learned a lot from him, and I am thankful for the time and effort he put into helping me.

Contents

Chapter 1

Introduction

This report investigates several notions of dimension for subsets of Euclidean space, with a particular emphasis on fractal geometry. The term *fractal* was introduced by Benoît B. Mandelbrot to describe sets that exhibit intricate structure and self-similarity at all scales. Classical geometric concepts are often inadequate to capture the complexity observed in natural objects such as snowflakes, floral patterns, and certain stochastic paths.

We present explicit examples of bounded subsets of $\mathbb{R}$ that, despite being uncountable, have zero length, as well as subsets of $\mathbb{R}^2$ with infinite perimeter but zero area. These examples motivate the introduction of alternative notions of dimension that more accurately quantify the geometric and analytic properties of such sets.

A summary of the key results is provided in Table 1.1. While many of these results are classical and well documented in the literature, rigorous proofs are often not readily accessible to learners. To solve that problem, we have provided independent proofs where appropriate, citing standard references when external results have been used. We have then applied these ideas to give a precise definition of fractals and discussed some interesting properties.

We will primarily work with compact sets in $\mathbb{R}$ and $\mathbb{R}^2$. Note that analysis of dimension for sets in general $\mathbb{R}^n$ has also been done extensively (see [Mun] and [Mat99]). The dimension of a set is formally thought of as the minimum number of coordinates needed to specify any point in it. Consequently one may view a line as a set of dimension one as any line in $\mathbb{R}^n$ passing through some point $\vec{x_0} \in \mathbb{R}^n$ in the direction of $\vec{x} \in \mathbb{R}^n$ can be parameterized as $L = \{\lambda\vec{x} + \vec{x_0} \mid \lambda \in \mathbb{R}\}$ and thus any point in L can be represented by one parameter $\lambda \in \mathbb{R}$. The interval $I = [0,1]$ (see Figure 1.1), is a set of length 1 and area 0. We can see that any point $x \in I$ can be represented by a single coordinate x and would view it as a "one-dimensional subset" of $\mathbb{R}$. Now the box $B = [0,1]^2 \subset \mathbb{R}^2$, (see Figure 1.2), has perimeter 4 and area 1. We can see that every point $(x,y) \in B$ can be represented by two coordinates $x, y \in \mathbb{R}$ and would view it as a "two-dimensional" subset of $\mathbb{R}^2$. Let us now consider the unit circle $S^1 := \left\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\right\}$ (see Figure 1.3). It has a perimeter of 2π and no area. It's standard to see that any point in S^1 can be parameterized as $(t, \pm\sqrt{1-t^2})$ where $t \in [-1,1]$ and it should have ”dimension” 1. In summary for each of the above examples the length, area and the classical notion of dimension do not seem to capture the geometric properties of the set.

In this article we shall introduce four notions of dimension, namely the topological, box, similarity and Hausdorff dimensions. These will help better explain geometric properties of arbitrary sets. Before defining the above dimensions precisely we shall first discuss four sets which are fundamental examples of fractals, namely the α–Cantor Set C^α, the Sierpiński Gasket S_3, the Sierpiński Carpet S_4 and finally the Koch Snowflake K. These sets are formally classified as fractals due to their due to their self-similar structure at all scales. We shall begin by defining C^α.

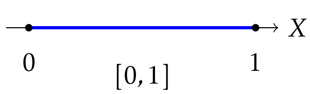


Figure 1.1: Interval $I \subset \mathbb{R}$

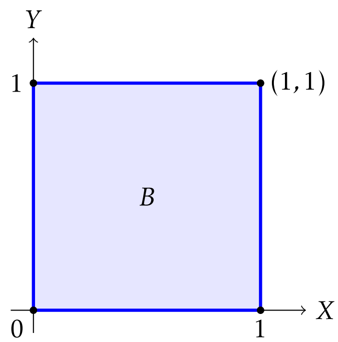


Figure 1.2: Box $B \subset \mathbb{R}^2$

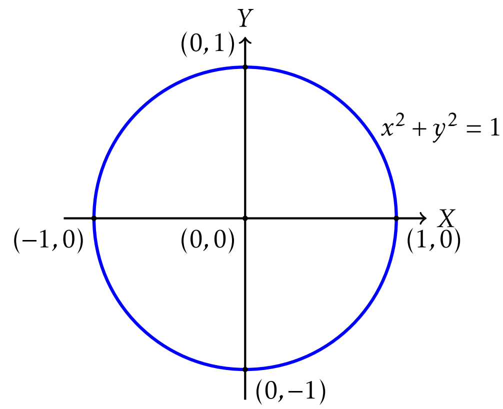


Figure 1.3: Circle $S^1 \subset \mathbb{R}^2$

Example 1. *(*α–**Cantor Set** C^α*) The Cantor set constructed by German mathematician George Cantor is a subset of* $[0,1]$. *Given* $\alpha \in (0,1)$, C^α *is defined recursively as follows. We start with the interval*

$$C_0 = [0,1].$$

We generate C_1 *by removing the middle open* $\alpha - th$ *segment from* C_0 *to get*

$$C_1 = \left[0, \frac{1-\alpha}{2}\right] \bigcup \left[\frac{1+\alpha}{2}, 1\right].$$

Next by removing the middle open $\alpha - th$ *segment from both intervals of* C_1 *we obtain*

$$C_2 = \left[0, \left(\frac{1-\alpha}{2}\right)^2\right] \bigcup \left[\frac{1-\alpha}{2} - \left(\frac{1-\alpha}{2}\right)^2, \ \frac{1-\alpha}{2}\right] \bigcup \left[\frac{1+\alpha}{2}, \ \frac{1+\alpha}{2} + \left(\frac{1-\alpha}{2}\right)^2\right] \bigcup \left[1 - \left(\frac{1-\alpha}{2}\right)^2, \ 1\right].$$

Suppose at the n^{th} *iteration the set generated* C_n *can be represented as a union of* 2^n *closed line segments. we remove the middle open* α^{th} *segment from each line segment as shown in Figure 1.4 to generate the* 2^{n+1} *closed line segments constituting* C_{n+1}.

Since we are removing line segments at every iteration, we know that

$$C_0 \supset C_1 \supset C_2 \dots$$

is a decreasing sequence of sets. We define our set

$$C^\alpha = \bigcap_{i=0}^{\infty} C_n.$$

$\square$

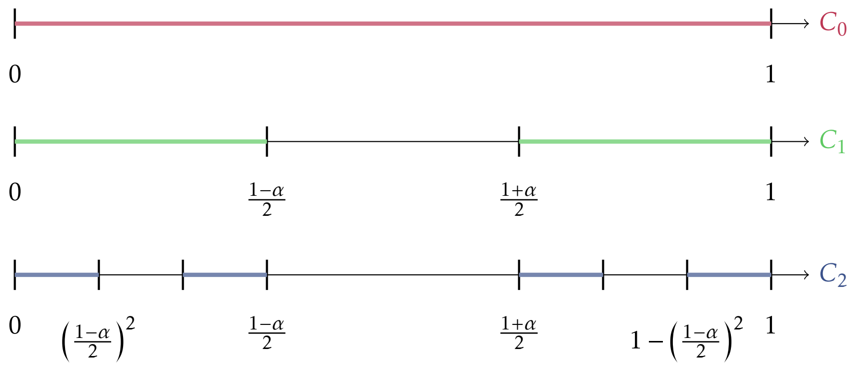


Figure 1.4: The First Three Iterations of C^α

We shall prove that C^α is an uncountable set (see Lemma 4.1.2) of length 0 (see Lemma 4.1.2), for all $\alpha \in (0,1)$.

We shall also prove an interesting property of the $\frac{1}{3}$–Cantor Set C. We shall show that the $C \subset [0,1]$ is constituted of exactly those points that can be represented by the digits 0 and 2 in ternary expansion (see Lemma 4.1.3).

We may think of a 0–dimensional object as a point and a one-dimensional object as a line with non-zero length. Clearly, C^α fits into neither and this is a motivation for exploring the four dimensions explored earlier. In our next example we will introduce a set designed by Polish mathematician Wacław Sierpiński that is generated by a triangle in $\mathbb{R}^2$.

Example 2. (Sierpiński Gasket $\mathbf{S_3}$) *We construct S_3 as follows. Let T_0 be the equilateral triangle of side length 1. Say T_n is the union of 3^n equilateral triangles of side length 2^{-n}, each pair sharing at most one vertex. In each triangle, we join the midpoints of the sides to obtain 4 congruent equilateral triangles and then remove the open inverted triangle. Doing this to all triangles yields 3^{n+1} equilateral triangles of side length $2^{-(n+1)}$, each pair sharing at most one vertex as shown in Figure 1.5. We define this set to be T_{n+1}. Since we are removing open triangles at every iteration, we know that*

$$T_0 \supset T_1 \supset T_2 \ldots$$

is a decreasing sequence of sets. Continuing this process gives us our limit set

$$S_3 = \lim_{n\to\infty} T_n = \bigcap_{n=0}^{\infty} T_n.$$

The R code for generating the Sierpiński Gasket will be available on request.

We shall prove that S_3 is a set of infinite perimeter (see Lemma 4.2.2) and zero area (see Lemma 4.2.2). We think of a one-dimensional object as a line with non-zero length and a two-dimensional object as a plane with non-zero area. Clearly, S_3 fits into neither and we cannot yet answer the question of its dimension much like the Cantor set. We will now discuss a fractal designed also by Sierpiński, generated by the box $B \subset \mathbb{R}^2$.

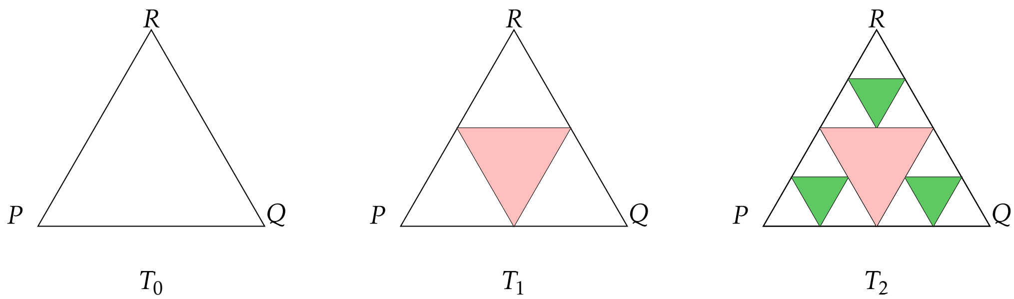


Figure 1.5: Sierpiński Gasket (S_3) at the first 3 levels.

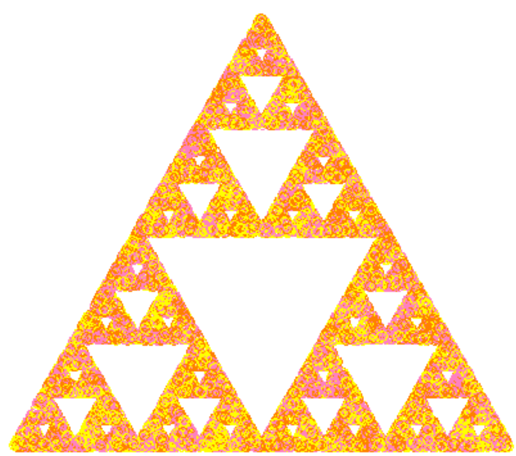


Figure 1.6: Sierpiński Gasket Generated on R

Example 3. **(Sierpiński Carpet S_4)** *Now that we have discussed the Sierpiński Gasket, as our third fractal of choice we will discuss the Sierpiński Carpet S_4. We construct S_4 as follows. Let $U_0 = [0,1] \times [0,1]$. We divide U_0 into 9 closed sub-boxes, i.e.,*

$$U_0 = \left\{ \left[0, \frac{1}{3}\right], \left[\frac{1}{3}, \frac{2}{3}\right], \left[\frac{2}{3}, 1\right] \times \left[0, \frac{1}{3}\right], \left[\frac{1}{3}, \frac{2}{3}\right], \left[\frac{2}{3}, 1\right] \right\}$$

and then remove the open middle sub-box $\left(\frac{1}{3}, \frac{2}{3}\right) \times \left(\frac{1}{3}, \frac{2}{3}\right)$ from U_0 to get

$$U_1 = \bigcup_{\substack{i,j \in \{0,1,2\} \\ (i,j) \neq (1,1)}} \left[\frac{i}{3}, \frac{i+1}{3}\right] \times \left[\frac{j}{3}, \frac{j+1}{3}\right].$$

Thus U_1 is a union of 8 closed sub-boxes. We repeat our process for each box in U_1 by dividing each box into 9 further sub-boxes each and then remove the open middle sub-box from each box in U_1 to get

$$U_2 = \bigcup_{\substack{i,j \in \{0,1,2\} \\ (i,j) \neq (1,1)}} \bigcup_{\substack{k,\ell \in \{0,1,2\} \\ (k,\ell) \neq (1,1)}} \left[\frac{3i+k}{9}, \frac{3i+k+1}{9}\right] \times \left[\frac{3j+\ell}{9}, \frac{3j+\ell+1}{9}\right].$$

If U_n is a union of 8^n closed boxes, we construct U_{n+1} by dividing each box in U_n into 9 sub-boxes each and removing the open middle sub-box from each box in U_n to get 8^{n+1} closed boxes.

Since $U_0 \supset U_1 \supset U_2 \ldots$, we define

$$S_4 = \bigcap_{n=0}^{\infty} U_n.$$

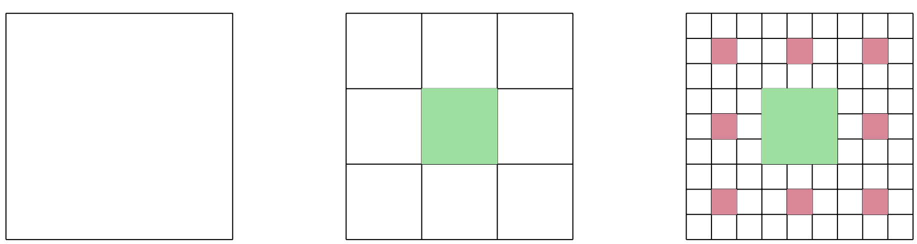


Figure 1.7: The first 3 iterations of S_4

The R code for generating the Sierpiński Carpet will be available on request.

We shall prove later that just like S_3, the perimeter of S_4 diverges to ∞ (see Lemma 4.3.2) and it has zero area (see Lemma 4.3.2). We may think of a one-dimensional object as a line with non-zero length and a two-dimensional object as a plane with non-zero area. Clearly, S_4 fits into neither and we shall answer the question of its dimension later. Lastly, we will introduce a set designed by Swedish mathematician Helge von Koch that is generated by the boundary of a triangle.

Example 4. **(Koch Snowflake K)** *K is constructed by starting with an equilateral triangle K_1 of side length 1. Given K_n, we get K_{n+1} by recursively altering each line segment of K_n as follows*

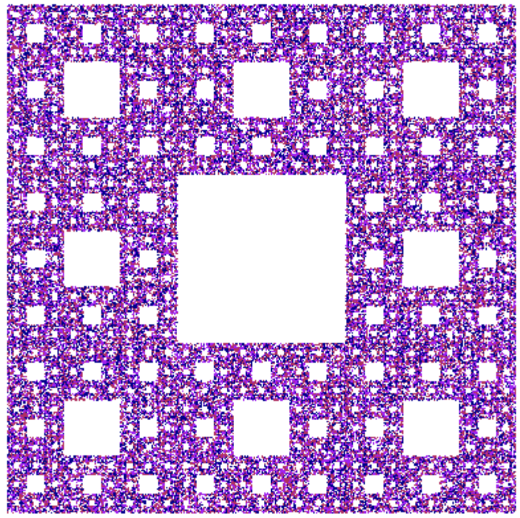


Figure 1.8: Sierpiński Carpet Generated on R

(see Figure 1.10). First we divide the line segment into three segments of equal length. Then we draw an equilateral triangle that has the middle segment as its base and points outward and then we remove the line segment that is the base of the triangle to get K_{n+1}. K cannot be represented as an intersection of these sets as the sets are not decreasing in nature. We will provide a precise definition later. For now we shall think of K as a limit set of the K_n's, the word limit being formal here.

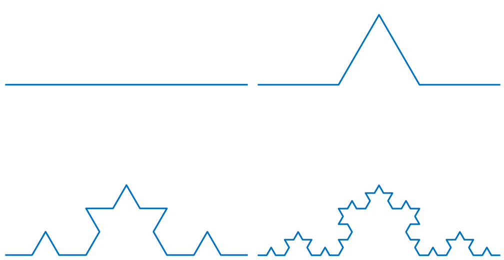


Figure 1.9: Construction of the Koch Curve [TIF25]

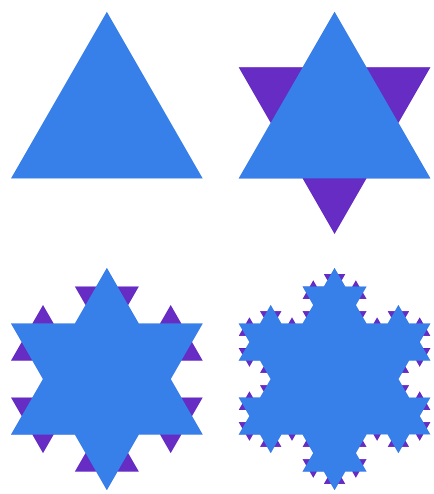


Figure 1.10: Construction of Koch Snowflake [TIF25]

We shall prove that K is a set of infinite perimeter (see Lemma 4.4.2) and zero area (see Lemma 4.4.2). We may think of a one-dimensional object as a line with non-zero length and a two-dimensional object as a plane with non-zero area. Clearly, K fits into neither and we shall answer the question of its dimension later.

We will demonstrate that the dimension of a set is not necessarily an integer. More strikingly, the concept of dimension is not always unique: its value can vary depending on the definition used. A dimension chart for the sets we have considered is given in Table 1.1.

Set	**Perimeter**	**Area**	**Topological**	**Box**	**Similarity**	**Hausdorff**
Interval I	1	0	1	1	1	1
Box B	4	1	2	2	2	2
Circle S^1	2π	0	1	1	–	1
α-Cantor Set C^α	0	0	0	$\log_{\frac{2}{1-\alpha}}(2)$	$\log_{\frac{2}{1-\alpha}}(2)$	$\log_{\frac{2}{1-\alpha}}(2)$
Sierpiński gasket S_3	∞	0	1	$\log_2(3)$	$\log_2(3)$	$\log_2(3)$
Sierpiński carpet S_4	∞	0	1	$\log_3(8)$	$\log_3(8)$	$\log_3(8)$
Koch snowflake K	∞	0	1	$\log_3(4)$	$\log_3(4)$	$\log_3(4)$

Table 1.1: A Summary of Results

The aim of this report is to calculate all the dimensions in Table 1.1 precisely. We will start with the non-fractal sets, i.e., the interval, box and circle and then move on to the fractals.

Layout of the report: The rest of the report is organized as follows. In the following section, we introduce three notions of dimension considered in this work - the topological, box and Hausdorff dimensions and discuss their properties. In section 3 we calculate the dimensions of the interval I (section 3.1), the box B (section 3.2) and the unit circle S^1 (section 3.1). Then in section 4 we formally define the fractals, namely the α–Cantor Set C^α (section 4.1), the Sierpiński gasket S_3 (section 4.2), the Sierpiński carpet S_4 (section 4.3) and the Koch snowflake K (section 4.4), discuss some interesting properties and then evaluate their dimensions. In the last section, we introduce the concept of similarity dimension and evaluate them for the above sets.

Chapter 2

Preliminaries

In this section we define three dimensions of a set - the topological, box and Hausdorff dimensions. We must cover some basic analytic and topological concepts before arriving at the definitions. Since we only deal with compact subsets of $\mathbb{R}$ and $\mathbb{R}^2$, we will define everything for subsets of $\mathbb{R}^n$ specifically.

Definition 2.0.1. **(Magnitude and Distance in $\mathbb{R}^n$)** *We will denote a vector $(x_1, x_2, \dots x_n) \in \mathbb{R}^n$ as x. Let $x \in \mathbb{R}^n$. The **magnitude** of x is given by*

$$\sqrt{\sum_{i=1}^{n} x_i^2}$$

*and is called the norm of x. Let $y \in \mathbb{R}^n$. The **distance** between x and y is given by*

$$\|x - y\| = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

□

Definition 2.0.2. **(Open balls and open sets)** *Let $X \subset \mathbb{R}^n$. For $p \in X$ and $r > 0$, an **open ball** $B(p, r) := \{y \in X \mid \|p - y\| < r\}$. X is said to be **open** in $\mathbb{R}^n$ if $\forall\ x \in X$, $\exists\ \epsilon_x > 0$ such that $B(x, \epsilon_x) \subset X$.* □

Lemma 2.0.3. *All open balls of $\mathbb{R}^n$ are open.*

Proof. Refer to Theorem 2.19 of [Rud64]. □

Definition 2.0.4. *(**Limit Point and Closure**) Given $X \subset \mathbb{R}^n$, x is said to be a **limit point** of X if $\forall\ \epsilon > 0$, $B(x, \epsilon) \cap X \supsetneq \{x\}$. The set of all limit points of X is denoted by X' and the **closure** of a set $\overline{X} := X \cup X'$.* □

Definition 2.0.5. *$X \subset \mathbb{R}^n$ is said to be **closed** in $\mathbb{R}^n$ if $\overline{X} \subset X$, i.e, X contains all its limit points.*

Lemma 2.0.6. *$X \subset \mathbb{R}^n$ is closed in $\mathbb{R}^n$ iff $X^c := \mathbb{R}^n \setminus X$ is open in $\mathbb{R}^n$.*

Proof. Refer to Theorem 2.23 of [Rud64]. □

Lemma 2.0.7. *The following hold for sets in* $\mathbb{R}^n$*:*

(a) Arbitrary unions and finite intersections of open sets are open.

(b) Finite unions and arbitrary intersections of closed sets are closed.

Proof. Refer to Theorem 2.24 of [Rud64]. $\square$

Definition 2.0.8. *(**Convergence**) Let* $X \subset \mathbb{R}^n$ *and* $\{x_n\}_{n\in\mathbb{N}}$ *be a sequence in* X*. Then we say that* ($\{x_n\}$ ***converges****/tends to* x *as* $n \to \infty$) *or* ($x_n \to x$ *as* $n \to \infty$) *or* ($\lim_{n\to\infty} x_n = x$) *if*

$$\forall\ \epsilon > 0,\ \exists\ N \in \mathbb{N} \text{ such that } \forall\ n > N,\ \|x_n - x\| < \epsilon.$$

$\square$

Lemma 2.0.9. *Given* $X \subset \mathbb{R}^n$, $x \in X$ *is a* ***limit point*** *of* X *iff* $\exists$ *a convergent sequence* $\{x_n\}_{n\in\mathbb{N}}$ *in* X *such that* $\lim_{n\to\infty} x_n = x$.

Proof. ($\Rightarrow$) Suppose x is a limit point of X. By the definition of a limit point, for every $\epsilon > 0$, the open ball $B(x,\epsilon) \cap X \supsetneq \{x\}$, i.e.,

$$\forall \epsilon > 0,\ \exists x_\epsilon \in X \setminus \{x\} \text{ such that } \|x_\epsilon - x\| < \epsilon.$$

We construct a sequence $\{x_n\} \subset X \setminus \{x\}$ by choosing, for each $n \in \mathbb{N}$, a point $x_n \in X \setminus \{x\}$ such that

$$\|x_n - x\| < \frac{1}{n}.$$

Then, by construction, $\lim_{n\to\infty} x_n = x$. Hence, such a sequence exists.

($\Leftarrow$) Conversely, suppose there exists a sequence $\{x_n\} \subset X \setminus \{x\}$ such that $\lim_{n\to\infty} x_n = x$. Then, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\|x_n - x\| < \epsilon,$$

i.e., $x_n \in B(x,\epsilon) \cap (X \setminus \{x\})$, so the intersection is non-empty. Thus, by the open ball definition, x is a limit point of X. $\square$

Definition 2.0.10 (**Basis on a Set** $X \subset \mathbb{R}^n$)**.** *Let* $X \subset \mathbb{R}^n$*. A collection* $\mathcal{B} = \{B_i\}_{i\in I}$ *of subsets of* X *is called a* ***basis*** *on* X *if it satisfies the following properties:*

1. *For each* $i \in I$*, there exists an open set* $U_i \subset \mathbb{R}^n$ *such that* $B_i = U_i \cap X$.
2. *For every point* $x \in X$*, there exists at least one set* $B_i \in \mathcal{B}$ *such that* $x \in B_i$.
3. *If* $x \in B_1 \cap B_2$ *for some* $B_1, B_2 \in \mathcal{B}$*, then there exists* $B_3 \in \mathcal{B}$ *such that* $x \in B_3 \subset B_1 \cap B_2$.

Equivalently, a collection $\mathcal{B} = \{\mathcal{B}_i\}_{i\in I} \subset X$ is a basis on $X \subset \mathbb{R}^n$ if for every point $x \in X$, and every set $U_x \subset X$ containing x that can be written as $U_x = U \cap X$ for some open set $U \subset \mathbb{R}^n$, there exists $\mathcal{B}_i \in \mathcal{B}$ such that $x \in \mathcal{B}_i \subset U_x$. $\square$

A standard basis for $\mathbb{R}^n$ is given by the collection of open balls with rational centers and radii:

$$\mathcal{B} = \{B(p,r) \,\big|\, p \in \mathbb{Q}^n,\ r \in \mathbb{Q}^+\},$$

where $B(p,r) := \{x \in \mathbb{R}^n : \|x - p\| < r\}$.

Definition 2.0.11. *(**Cover and Refinement**) Let $X \subset \mathbb{R}^n$. Then the set of open sets $\mathcal{U} := \{\mathcal{U}_i\}_{i \in I} \subset \mathbb{R}^n$ is said to be an **open cover** of X if $X \subset \bigcup_{i \in I} \mathcal{U}_i$.*
*A subset of $\mathcal{U}$ that covers X is known as a **sub-cover** of $\mathcal{U}$.*
*A **refinement** $\mathcal{V} = \left\{\mathcal{V}_\beta\right\}_{\beta \in B}$ of $\mathcal{U}$ is an open cover of X such that $\forall\ \beta \in B$, $\exists\ \alpha \in A$ such that $\mathcal{V}_\beta \subset \mathcal{U}_\alpha$.* □

Definition 2.0.12. *(**Boundedness and Compactness**) $X \subset \mathbb{R}^n$ is said to be **bounded** if $\exists\ M > 0$ such that $\|x\| < M\ \forall\ x \in X$.*
*$X \subset \mathbb{R}^n$ is said to be **compact** if every open cover of X admits a finite sub-cover.* □

Theorem 2.0.13. Heine-Borel: *For a subspace $S \subset \mathbb{R}^n$, the following are equivalent:*

1. *S is closed and bounded*
2. *S is compact*

Proof. Refer to Theorem 2.41 of [Rud64]. □

Lemma 2.0.14. *Let A, B be disjoint compact subsets in $\mathbb{R}^n$. Then*

$$d(A,B) := \inf\{\|a - b\| : a \in A,\ b \in B\} > 0.$$

Proof. Refer to [pro22]. □

Definition 2.0.15. *(**Continuous Map**) Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$. A map $f : X \to Y$ is said to be continuous if $f^{-1}(V) \subset X$ is open(closed) in $\mathbb{R}^n$ for all $V \subset Y$ open(closed) in $\mathbb{R}^m$. This definition can be translated as follows:*

$$\forall\ x \in \mathbb{R}^n,\ \forall\ \epsilon > 0,\ \exists\ \delta > 0 \text{ such that } \|f(x) - f(y)\| < \epsilon\ \forall\ \|x - y\| < \delta.$$

□

Definition 2.0.16. *(**Homeomorphism**) Given $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$, a **homeomorphism** is a map $f : X \mapsto Y$ such that:*

1. *f is bijective*
2. *f is continuous*
3. *$f^{-1} : Y \to X$ is continuous*

Lemma 2.0.17. *A homeomorphism is both open and closed in nature, i.e., it maps open(closed) sets to open(closed) sets and also pulls back open(closed) maps to open(closed) maps.*

Proof. Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be homeomorphic, i.e., $\exists\ f : X \to Y$, a homeomorphism. It is clear from Definition 2.0.15 and Definition 2.0.16 that f pulls back open(closed) sets to open(closed) sets due to continuity. We will show that f is both an open and closed map.

Open map: Let $U \subset X$ and $V = f(U)$. Since f is bijective, $f^{-1}(V) = U$ which we know is open in X. Since f^{-1} is continuous, $\left(f^{-1}\right)^{-1}(U) = V$ is open in Y. Therefore, f maps open sets to open sets.

Closed map: Let $C \subset X$ be closed. Then $X \setminus C$ is open in X, so $f(X \setminus C)$ is open in Y (by openness of f). But

$$f(X \setminus C) = f(X) \setminus f(C) = Y \setminus f(C)$$

because f is bijective. Hence, $f(C)$ is closed in Y. Therefore, f maps closed sets to closed sets. □

Definition 2.0.18. *(**Connectedness and Path Connectedness**) $X \subset \mathbb{R}^n$ is said to have a **separation** if $\exists$ non-empty sets A, B open in $\mathbb{R}^n$ such that $(A \cap X) \bigsqcup (B \cap X) = X$, where $\bigsqcup$ is the disjoint union. $X_1, X_2 \subset X$ are said to be **separated** in X if $\exists$ A, B as above such that $X_1 \subset A$ and $X_2 \subset B$. X is said to be **connected** if it has no separation.*

*X is said to be **totally disconnected** if any two non-empty subsets of X are separated.*

*Given $x_1, x_2 \in X \subset \mathbb{R}^n$, we say that there is a **path** from x_1 to x_2 in X if $\exists$ a continuous function $\gamma : [0,1] \to X$ such that $\gamma(0) = x_1$ and $\gamma(1) = x_2$. X is said to be **path connected** if there exists a path in X between any two points.* $\square$

Lemma 2.0.19. *Every path connected set is connected.*

Proof. Refer to Chapter 3 of [Mun]. $\square$

Lemma 2.0.20. *Let $\{A_i\}_{i \in I} \subset \mathbb{R}^n$ be connected sets such that $\nexists$ $i \in I$ such that $A_i \bigcap \bigcup_{i \neq j \in I} A_j = \emptyset$. Then $\bigcup_{i \in I} A_i$ is connected in $\mathbb{R}^n$.*

Proof. Refer to theorem 23.3 of [Mun]. $\square$

We will start with defining the topological dimension of a set which is an integral dimension. This generalizes the concept of the dimension of a vector space to any set.

Definition 2.0.21 (Lebesgue Covering Dimension (Topological Dimension)). *Let $X \subset \mathbb{R}^n$, and let $\mathcal{V} = \{\mathcal{V}_i\}_{i \in I}$ be an open cover of X. For each point $x \in X$, define*

$$N_x := \#\{i \in I \mid x \in \mathcal{V}_i\},$$

*i.e., the number of elements of $\mathcal{V}$ that contain x. The **order** of $\mathcal{V}$ is defined as*

$$\operatorname{ord}(\mathcal{V}) := \sup\{N_x \mid x \in X\}.$$

*The **topological dimension** of X is defined by*

$$\dim_T(X) := \min\{d \geq 0 \mid \textit{every open cover } \mathcal{U} \textit{ of } X \textit{ has an open refinement } \mathcal{V} \textit{ with } \operatorname{ord}(\mathcal{V}) \leq d+1\}.$$

We define $\dim_T(\emptyset) := -1$. For any nonempty space, the topological dimension is either a non-negative integer or ∞. $\square$

The topological dimension of a set happens to be scale invariant. If we scale a set $X \subset \mathbb{R}^n$ by some constant $c \geq 0$, we can always scale the sets in our open cover of X by the same c to cover cX. Here we have only dealt with compact subsets of $\mathbb{R}^n$. Using Lemma 2.0.24, it is trivial to provide an upper bound for the topological dimension of all the sets considered. However for further clarity, while proving that a set has topological dimension less than or equal to d, we have tried to construct explicit refinements of order $d+1$ given arbitrary open covers in order to place the upper bound. In order to do this, we use the Lebesgue Covering Lemma.

Definition 2.0.22. *(**Lebesgue Number**) The **diameter** of a set $X \subset \mathbb{R}^n$ is given by*

$$diam(X) := \sup\{\|x - y\| \mid x, y \in X\}.$$

*Let $\mathcal{U} = \{\mathcal{U}_i\}_{i \in I}$ be an open cover of X. Then the **Lebesgue Number** of $\mathcal{U}$,*

$$\delta := \sup\{s \geq 0 \mid \forall\ Y \subset X \textit{ satisfying } diam(Y) \leq s,\ \exists\ i \in I \textit{ such that } Y \subset \mathcal{U}_i\}.$$

$\square$

Lemma 2.0.23. *(**Lebesgue Covering Lemma**) Any open cover of a compact subset of $\mathbb{R}^n$ has a positive Lebesgue Number.*

Proof. Refer to Lemma 27.5 of [Mun]. $\square$

Lemma 2.0.23 is useful in providing an upper bound for the topological dimension of a compact set. Say $X \subset \mathbb{R}^n$ is compact. In order to prove that a X has topological dimension lesser than or equal to d, we start with any arbitrary open cover $\mathcal{U}$ of X. By Lemma 2.0.23, $\mathcal{U}$ must have a Lebesgue number $\delta > 0$. By the definition of Lebesgue number, all sets of diameter lesser than or equal to δ will be contained in some set of $\mathcal{U}$. It is then enough to construct an open cover $\mathcal{V}$ of X of order $d+1$ with sets that have diameter lesser than or equal to δ. Then all sets of $\mathcal{V}$ will be contained in some set of $\mathcal{U}$ and $\mathcal{V}$ will act as a refinement of $\mathcal{U}$ of order $d+1$. Since we can do this for any arbitrary open cover, by definition of topological dimension, the topological dimension of X must be at most d. We now state some properties of topological dimension below.

Theorem 2.0.24. *Given $n \in \mathbb{N}$, every compact subset of $\mathbb{R}^n$ has topological dimension at most n.*

Proof. Refer to Theorem 50.6 of [Mun]. $\square$

Lemma 2.0.25. *Let $X \subset Y \subset \mathbb{R}^n$. Then $\dim_T(X) \le \dim_T(Y)$, i.e., topological dimension is monotonic under inclusion.*

Proof. Refer to Theorem 50.1 of [Mun]. $\square$

Lemma 2.0.26. *$X, Y \subset \mathbb{R}^n$ be homeomorphic, i.e., $\exists\ f : X \to Y$, a homeomorphism. Then their topological dimensions $\dim_T(X) = \dim_T(Y)$.*

Proof. We will show that $\dim X \le n$ if and only if $\dim Y \le n$, which implies $\dim X = \dim Y$. Assume $\dim X \le n$. Let $\mathcal{V}$ be an open cover of Y. Since f is a homeomorphism, the family

$$\mathcal{U} = \{f^{-1}(V) : V \in \mathcal{V}\}$$

is an open cover of X. By assumption, there exists an open refinement $\mathcal{U}'$ of $\mathcal{U}$ of order at most $n+1$. Then for all $U' \in \mathcal{U}'$, $\exists\ U \in \mathcal{U}$ such that $U' \subset U$. By definition of $\mathcal{U}$, for all $U' \in \mathcal{U}'$, $\exists\ V \in \mathcal{V}$ such that $U' \subset f^{-1}(V)$. Note that since f is bijective,

$$U' \subset f^{-1}(V) \iff f(U') \subset V.$$

Then the family

$$\mathcal{V}' = \{f(U') : U' \in \mathcal{U}'\}$$

is an open cover of Y and is a refinement of $\mathcal{V}$. Since $\mathcal{U}'$ is of order at most $n+1$,

$$\sup_{x \in X} \#\{i \in I : x \in U'_i\} \le n+1.$$

Let $y \in Y$. Since f is a bijection,

$$\#\{i \in I : y \in f(U'_i)\} = \#\{i \in I : f^{-1}(y) \in U'_i\} \le n+1.$$

Since $f^{-1}(Y) = X$, taking supremum over all $y \in Y$, we conclude:

$$\operatorname{ord}(\mathcal{V}') = \sup_{y \in Y} \#\{i \in I : y \in f(U'_i)\} = \sup_{x \in X} \#\{i \in I : x \in U'_i\} \le n+1.$$

This shows that $\mathcal{V}'$ is a refinement of $\mathcal{V}$ of order at most $n+1$. Therefore, every open cover $\mathcal{V}$ of Y has an open refinement of order $\leq n+1$, so $\dim_T(Y) \leq n$.

The reverse implication $\dim Y \leq n \Rightarrow \dim X \leq n$ follows by applying the same argument to the inverse map $f^{-1} : Y \to X$, which is also a homeomorphism.

Therefore, $\dim X = \dim Y$. $\square$

Lemma 2.0.27. *Let $X \subset \mathbb{R}^n$ be compact. Then X is totally disconnected in $\mathbb{R}^n \iff X$ has a basis of clopen sets.*

Proof. Refer to Chapter IX of [Joh82]. $\square$

Theorem 2.0.28. *The topological dimension of a non-empty compact subset of $\mathbb{R}^n$ is 0 $\iff$ it is a totally disconnected set.*

Proof. $\implies$ Let $X \subset \mathbb{R}^n$ compact, have topological dimension 0. Then, by definition, every open cover $\mathcal{U}$ has a refinement $\mathcal{V}$ of order 1, i.e., its elements are disjoint sets. Say $\mathcal{V} = \{\mathcal{V}_i\}_{i\in I}$. Notice that $\forall\ i \in I,\ V_i^c = \bigcup_{i\neq j\in I} V_j$ is an union of open balls $\implies V_i^c$ is open $\implies V_i$ is closed $\implies V_i$ is clopen.

Let $x, y \in X$. Consider $U_x = B\left(x, \frac{\|x-y\|}{2}\right)$ and $U_y = B\left(y, \frac{\|x-y\|}{2}\right)$. Then $\exists$ disjoint, open $U_x \ni x$, $U_y \ni y$. Let $\mathcal{U}$ be an open cover of X containing U_x and U_y. It then has a refinement $\mathcal{V}$ of disjoint clopen sets such that x and y lie in disjoint sets of $\mathcal{V}$, say $x \in \mathcal{V}_x$. $\mathcal{V}_x$ is clopen as shown above $\implies y \in \mathcal{V}_x^c$ is also clopen. Hence, $\forall\ x, y \in X$, we have a separation of x and y given by $\mathcal{V}_x$ and $\mathcal{V}_x^c$. Therefore X is totally disconnected.

$\impliedby$ Let $X \subset \mathbb{R}^n$ compact, be totally disconnected. By 2.0.27, X has a basis of clopen sets, say $W = \{W_i\}_{i\in I}$. Let $\mathcal{U}$ be a cover of X. Since X is compact, $\mathcal{U}' = \{\mathcal{U}_1, \mathcal{U}_2, \ldots \mathcal{U}_m\}$ is a finite sub-cover of $\mathcal{U}$. $\forall\ x \in X$, we choose $W_x \in W$ such that $x \in W_x \subset U_x$ where $U_x \in \mathcal{U}'$.

Thus $\{W_x\}_{x\in X}$ forms a clopen cover of X and due to compactness of X, has a finite sub-cover $\{W_1, W_2, \ldots, W_k\}$. Since all these sets are contained in some U_x, it is a refinement of $\mathcal{U}$. We now make these sets disjoint by considering the refinement

$$\mathcal{W} = \left\{ W_1, W_2 \setminus W_1, W_3 \setminus (W_1 \cup W_2), \ldots, W_k \setminus \bigcup_{i=1}^{k-1} W_i \right\},$$

which are also clopen as W is a clopen cover of X. Therefore, we get a refinement of disjoint clopen sets $\implies \dim_T(X) = 0$. $\square$

Lemma 2.0.29. *Let $\{A_i\}_{i\in I} \subset \mathbb{R}^n$ be a countable family of closed sets with finite topological dimension. Then:* $\dim\left(\bigcup_{i\in I} A_i\right) = \sup_{i\in I}\{\dim_T(A_i)\}$

Proof. Refer to Theorem 50.1 of [Mun]. $\square$

Lemma 2.0.30. *Given $X \subset \mathbb{R}^n$,* $\dim_T(X \times [0,1]) = \dim_T(X) + 1$.

Proof. Refer to corollary 3.52 of [KP07]. $\square$

Now that we have discussed the notion of topological dimension, we come to the real-valued dimensions. We start with the box-counting dimension which requires us to cover our set with small boxes of size $\epsilon > 0$ and counting the minimum number of boxes needed to do so.

Definition 2.0.31. *(***Box in** $\mathbb{R}^n$*) A set $B \subset \mathbb{R}^n$ is said to be a* ***closed box*** *if there exists an orthonormal linear map $R : \mathbb{R}^n \mapsto \mathbb{R}^n$, a translation vector $x_0 \in \mathbb{R}^n$ and $\{a_i\}_{i=1}^n \subset \mathbb{R}, \left\{b_j\right\}_{j=1}^n \subset \mathbb{R}$ such that*

$$B = R\left(\prod_{i=1}^{n} [a_i, b_i]\right) + x_0,$$

where $\prod$ is the usual Cartesian product. We shall define the ***size of such a box*** *to be*

$$\max\{| R(a_i - b_i) |: 1 \le i \le n\}.$$

$\square$

Definition 2.0.32. *(***The Minkowski-Bouligand Dimension or Box Dimension***)*

Let $X \subset \mathbb{R}^n$. Let $\epsilon > 0$ be given. Then $N(\epsilon)$ is the minimum number of boxes of size ϵ required to cover X. Then the ***box dimension*** *of X,*

$$\dim_B(X) := \lim_{\epsilon \to 0} -\frac{\log N(\epsilon)}{\log \epsilon}, \tag{2.1}$$

whenever the limit exists. If the limit does not exist, then we define the ***upper-box dimension*** *to be*

$$\overline{\dim}_B(X) := \limsup_{\epsilon \to 0} -\frac{\log(N(\epsilon)}{\log \epsilon}, \tag{2.2}$$

and the ***lower-box dimension*** *to be*

$$\underline{\dim}_B(X) := \liminf_{\epsilon \to 0} -\frac{\log(N(\epsilon)}{\log \epsilon}. \tag{2.3}$$

$\square$

It is easily observed that the box-dimension, upper-box dimension and lower-box dimension are non-negative real numbers and could be infinity.

Also note that when a set $X \subset \mathbb{R}^n$ is scaled by some constant $c \ge 0$,

$$\dim_B(cX) = \lim_{\epsilon \to 0} -\frac{\log(c \cdot N(\epsilon))}{\log \epsilon} = \lim_{\epsilon \to 0} -\frac{\log c + \log N(\epsilon)}{\log \epsilon}$$

where $N(\epsilon)$ is the minimum number of boxes of size ϵ required to cover X.

As $\epsilon \downarrow 0$, $\frac{\log c}{\log \epsilon} \to 0$. Thus

$$\dim_B(cX) = \lim_{\epsilon \to 0} -\left(\frac{\log c}{\log \epsilon} + \frac{\log N(\epsilon)}{\log \epsilon}\right) = \dim_B(X).$$

This shows that the box dimension of a set is scaling invariant.

Now we move on to our last dimension, namely the Hausdorff dimension. The Hausdorff dimension is a measure of *roughness* that was introduced by German mathematician Felix Hausdorff. In order to study this dimension, we require the concept of Borel measure.

Definition 2.0.33. **(Borel Set and Borel Measure)** *A* ***Borel subset*** *$\mathcal{B} \subset \mathbb{R}^n$ is any set that can be generated by the operations of countable union, countable intersection and relative complement on all open sets of $\mathbb{R}^n$. A* ***Borel measure*** *$\mu : \mathcal{B}(\mathbb{R}^n) \to \mathbb{R}$ is a map defined on $\mathcal{B}(\mathbb{R}^n)$, the set of Borel subsets of $\mathbb{R}^n$ satisfying:*

(a) ***Non-Negativity:*** *$\forall\ E \subset \mathcal{B}(\mathbb{R}^n)$, $\mu(E) \geq 0$.*

(b) ***Countable Additivity:*** *For all countable collection of disjoint sets $\{E_i\}_{i\in I} \in \mathcal{B}(\mathbb{R}^n)$,*

$$\mu\left(\bigcup_{i\in I} E_i\right) = \sum_{i\in I} \mu(E_i).$$

□

Definition 2.0.34. *(**Hausdorff Besicovitch Dimension**) Let $X \subset \mathbb{R}^n$ and $\delta > 0$. For $d > 0$,*

$$H_\delta{}^d(X) := \inf \left\{ \sum_i (diam(\mathcal{U}_i))^d : \{\mathcal{U}_i\}_i \text{, countable open cover of } X \text{ such that } diam(\mathcal{U}_i) \leq \delta\ \forall\ i \right\}.$$

From the above definition, we define the **d** *–* ***Hausdorff Measure*** *as*

$$H^d(X) := \lim_{\delta \to 0} H_\delta{}^d(X).$$

*Then the **Hausdorff dimension***

$$\dim_H(X) := \inf \left\{ d \geq 0 : H^d(X) = 0 \right\} = \sup \left\{ d \geq 0 : H^d(X) = \infty. \right\} \tag{2.4}$$

It is a non-negative real number and can be infinity. □

If d is the Hausdorff dimension of X, its s–Hausdorff Measure jumps from ∞ to 0 at $s = d$ as shown in Figure 2.1. The Hausdorff dimension is scale invariant as given a set

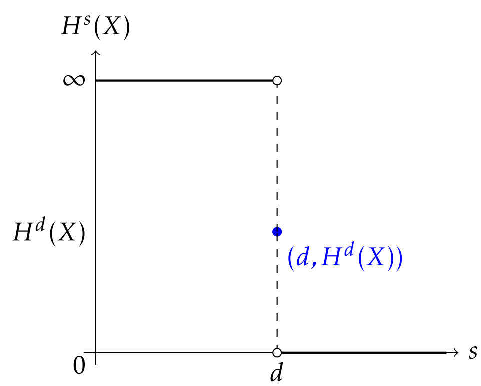


Figure 2.1: s–Hausdorff Measure of a set X of Hausdorff Dimension d

$X \subset \mathbb{R}^n$ and some $s \geq 0$, $H^s(X) = 0 \iff H^s(cX) = c^s H^s(X) = 0$ by definition of s–Hausdorff Measure. We will use a lemma called the Frostman's lemma to obtain a lower bound for the Hausdorff dimension of a set. The lemma is stated below

Lemma 2.0.35. *(**Frostman's Lemma**) The statements are equivalent for a Borel set $A \subset \mathbb{R}^n$:*

1. *$H^s(A) > 0$, where H^s denotes the s-Hausdorff measure.*

2. *There exists an (unsigned) Borel measure μ on $\mathbb{R}^n$ and a constant $c > 0$ satisfying $\mu(A) > 0$ such that $\mu(B(x,r)) \leq c \cdot r^s \ \forall \ x \subset \mathbb{R}^n$ and $r > 0$.*

Proof. Refer to Mass distribution principle 4.2 of [Fal13] and Theorem 8.8 of [Mat99]. □

This lemma is used to obtain a lower bound of a set. Say $X \subset \mathbb{R}^n$. In order to show that $\dim_H(X) \geq s$, it is enough to show that $H^s(X) > 0$ which is equivalent to finding a measure μ satisfying the required conditions of Frostman's Lemma. In the next section we calculate the dimensions of the standard sets, namely I, B and S^1.

Chapter 3

Dimensions of I, B and S^1

In this section, we calculate the topological, box and Hausdorff dimensions of the interval I, the box B and the circle S^1. We will start with the interval I.

3.1 Interval $\mathbf{I} = [\mathbf{0}, \mathbf{1}]$

We will show that the topological, box and Hausdorff dimensions of I are all equal to 1. We formally state this in the main result below.

Theorem 3.1.1. *Let* $I = [0, 1]$. *Then*

(a) $\dim_T(I) = 1$,

(b) $\dim_B(I) = 1$ *and*

(c) $\dim_H(I) = 1$.

3.1.1 Proof of Theorem 3.1.1(a)

We will show that the topological dimension $\dim_T(I) = 1$. In order to do this we will show first that $\dim_T(I) \geq 1$ and subsequently we will show $\dim_T(I) \leq 1$.

Proof of $\dim_T(I) \geq 1$. By Lemma 2.0.28, it is enough to show that I is compact and connected in order to prove this. We will show that $I \subset \mathbb{R}$ is compact using Theorem 2.0.13.

$[0, 1] \subset (-2, 2) = B(0, 2)$ readily implies I is bounded. We will next show that $I^c = (-\infty, 0) \cup (1, \infty)$ is open in $\mathbb{R}$. Let $x \in I^c$ be arbitrary. We will consider two cases, namely $x \in (-\infty, 0)$ and $x \in (1, \infty)$.

Case 1 $x \in (-\infty, 0)$*:* Let $\epsilon_0 = \frac{|x|}{2}$. Then note that $B(x, \epsilon_0) = (x - \epsilon_0, x + \epsilon_0) \subset (-\infty, 0)$.

Case 2 $x \in (1, \infty)$*:* Let $\epsilon_1 = \frac{x-1}{2}$. Then note that $B(x, \epsilon_1) = (x - \epsilon_1, x + \epsilon_1) \subset (1, \infty)$.

Therefore, we have shown that for all $x \in I^c$, $\exists\ \epsilon > 0$ such that $B(x, \epsilon) \subset I^c$. Thus I^c is open and I is closed. Thus we have shown that $[0, 1]$ is compact by Theorem 2.0.13 as $I \subset \mathbb{R}$. Also, I is an interval which implies it is path-connected and thus connected by Lemma 2.0.19. Thus I is a compact and connected subset of $\mathbb{R}$ and using Lemma 2.0.28, the topological dimension of I is greater than 0. Since the topological dimension is always an integer, we have $\dim_T(I) \geq 1$. □

We will now show that $\dim_T(I) \le 1$.

Proof of $\dim_T(I) \le 1$. In order to prove this, given any open cover $\mathcal{U} = \{\mathcal{U}_i\}_{i\in I}$, we have to construct a refinement of order 2. We use the concept of Lebesgue Number in order to achieve this. Let $\mathcal{U}$ be an open cover of I. Since I is compact, $\mathcal{U}$ has a positive Lebesgue number $\delta > 0$. Let $0 < \delta_0 < \min\{\delta, 1\}$ be such that $\frac{1}{\delta_0} = \frac{n}{2}$ for some $n \in \mathbb{N}$. We define

$$B_i = (i\delta_0, (i+1)\delta_0),$$

$\forall\ i \in J = \left\{-\frac{1}{2}, 0, \frac{1}{2}, \dots, \frac{1}{\delta_0} - \frac{1}{2}, \frac{1}{\delta_0}\right\}$ as in Figure 3.1.

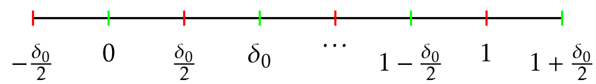


Figure 3.1: Refinement of Order 2

We need the following lemma to arrive at our result which we will assume for now. The proof to the lemma will be provided subsequently.

Lemma 3.1.2. *The set of sets $\{B_i\}_{i\in J}$ forms an open cover of $[0,1]$ of order 2.*

Assuming this lemma, since each B_i has diameter $\delta_0 < \delta$, by the definition of Lebesgue number this means that each B_i is contained in some set of $\mathcal{U}$ and thus $\{B_i\}_{i\in J}$ is a refinement of $\mathcal{U}$ of order two. Since our open cover $\mathcal{U}$ was arbitrary, we have shown that $\dim_T(I) \le 1$. □

Since we have shown that $1 \le \dim_T(I) \le 1$ we have $\dim_T(I) = 1$.

Proof of Lemma 3.1.2. We will prove the lemma in three steps.

Step 1 $B_i's$ **are open:** By definition, $B_i = B\left(\left(i + \frac{1}{2}\right)\delta_0, \frac{\delta_0}{2}\right)\ \forall\ i \in J$, which are open balls in $\mathbb{R}$, and are thus open.

Step 2 $B_i's$ **cover** I**:** Let $x \in [0,1]$. Take $i = \frac{1}{2}\left\lfloor \frac{2x}{\delta_0} \right\rfloor \in J$. It is easy to see that $x \in B_i$. Since $x \in [0,1]$ was arbitrary, $\{B_i\}_{i\in J}$ covers $[0,1]$.

Step 3 **The cover is of order 2:** Let $x \in [0,1]$. If $\frac{x}{\delta_0} \in J$, then $x \in B_i$ with $i = \frac{x}{\delta_0} - \frac{1}{2}$. Otherwise if $\frac{x}{\delta_0} \notin J$, then $x \in B_i$ where $i = \frac{1}{2}\left\lfloor \frac{2x}{\delta_0} \right\rfloor$ and $i = \frac{1}{2}\left\lceil \frac{2x}{\delta_0} - 2 \right\rceil$. Thus, each x lies in at most two $B_i's$ which implies that the cover $\{B_i\}_{i\in J}$ has order two.

□

3.1.2 Proof of Theorem 3.1.1(b)

A good visual understanding of this proof is given in [FU16]. By Definition 2.0.32, the box dimension of I is given by

$$\lim_{\epsilon \downarrow 0} -\frac{\log N(\epsilon)}{\log \epsilon}$$

where $N(\epsilon)$ is the minimum number of boxes in $\mathbb{R}$ of size ϵ required to cover $[0,1]$.

Let $\epsilon > 0$ be given. A box of size ϵ in $\mathbb{R}$ must be an interval of length ϵ. We claim that $N(\epsilon) = \left\lceil \frac{1}{\epsilon} \right\rceil$.

Suppose not, say there are $n < \left\lceil \frac{1}{\epsilon} \right\rceil$ boxes that cover I. Note that the total length covered by the n boxes is $n \cdot \epsilon < \frac{1}{\epsilon} \cdot \epsilon = 1$. Thus they cannot cover $[0,1]$. This shows that the number of boxes required must be at least $\left\lceil \frac{1}{\epsilon} \right\rceil$.

For any $x \in [0,1]$ note that $x \in \left[\left\lfloor \frac{x}{\epsilon} \right\rfloor \epsilon, \left(\left\lfloor \frac{x}{\epsilon} \right\rfloor + 1 \right) \epsilon \right]$. So

$$[0,1] \subset \bigcup_{i=0}^{\left\lceil \frac{1}{\epsilon} \right\rceil - 1} [i\epsilon, (i+1)\epsilon].$$

Therefore we have constructed a cover of $\left\lceil \frac{1}{\epsilon} \right\rceil$ sets and so the claim that $N(\epsilon) = \left\lceil \frac{1}{\epsilon} \right\rceil$ holds. Note that $\exists\ c \geq 1$ such that $\left\lceil \frac{1}{\epsilon} \right\rceil = c \cdot \frac{1}{\epsilon}$. Now

$$\frac{-\log N(\epsilon)}{\log \epsilon} = \frac{\log c - \log \epsilon}{-\log \epsilon} = -\frac{\log c}{-\log \epsilon} + 1.$$

As $\epsilon \downarrow 0$, we know that $\frac{\log c}{-\log \epsilon} \to 0$. Thus $\dim_B(I)$ exists and is equal to 1.

$\square$

3.1.3 Proof of Theorem 3.1.1(c)

By Definition 2.0.34, we need $d \geq 0$ such that the Hausdorff measure

$$H^s(I) = 0\ \forall\ s > d \text{ and } H^s(I) = \infty\ \forall\ s > d.$$

We will show that the Hausdorff Dimension of I is 1. In order to show this, we will first show that $\dim_H(I) \leq 1$.

Proof of $\dim_H(I) \leq 1$. This is equivalent to proving that $H^s(I) = 0\ \forall\ s > 1$. Let $s > 1$ be arbitrary. Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of I, $\mathcal{U} = \{\mathcal{U}_i\}_{i \in I}$ satisfying $\operatorname{diam}(\mathcal{U}_i) \leq \delta\ \forall\ i \in I$, with I being a suitable index set. By Definition 2.0.34,

$$H^s(I) = \lim_{\delta \downarrow 0} \inf_{\mathcal{U} \in \mathcal{P}_\delta} \sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s$$

where $\sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n \to \infty} \sum_{\substack{i \in I_n \\ |I_n| < \infty \\ I_n \uparrow I}} (\operatorname{diam}(\mathcal{U}_i))^s$ if I is countably infinite.

Let $0 < \delta < 1$ be fixed. Define $\delta_0 = \frac{2}{\left\lceil \frac{2}{\delta} \right\rceil}$. Then $0 < \delta_0 \leq \delta$ is such that $\frac{1}{\delta_0} = \frac{n}{2}$ for some $n \in \mathbb{N}$. Consider an open cover $B = \left\{ B_j \right\}_{j \in J}$ of I such that

$$B_j = (i\delta_0, (i+1)\delta_0)\ \forall\ j \in J = \left\{ -\frac{1}{2}, 0, \frac{1}{2}, \ldots, \frac{1}{\delta_0} - \frac{1}{2}, \frac{1}{\delta_0} \right\}.$$

Then the $\operatorname{diam}(B_i) = \delta_0 < \delta \ \forall \ i \in J$ and the minimum number of sets in B needed to cover I is $|J| = \frac{2}{\delta_0} + 2 = \left\lceil \frac{2}{\delta} \right\rceil + 2$. Since $B \in \mathcal{P}_\delta$,

$$\inf_{\mathcal{U} \in \mathcal{P}_\delta} \sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s \leq \sum_{j \in J} (\operatorname{diam}(B_j))^s = |J| \cdot \delta^s \leq \frac{2}{\delta} \cdot \delta^s + 2 \cdot \delta^s.$$

As $s > 1$, the right hand side of the above converges to zero as $\delta \downarrow 0$. From definition and an application of the squeeze theorem we have that

$$H^s(I) = 0.$$

Therefore $\dim_H(I) \leq 1$. □

We will now prove that $\dim_H(I) \geq 1$.

Proof of $\dim_H(I) \geq 1$. It is enough to prove that $H^1(I) > 0$. We use Frostman's Lemma in order to show this. We define a map $\mu : \mathcal{B}(I) \to [0,1]$ as the uniform distribution on $[0,1]$ given by:

$$\mu([a,b]) = \int_a^b 1 \ dx \ = \ b - a \ \ \forall \ 0 \leq a \leq b \leq 1.$$

For $x \in I$ it is easy to deduce that

$$\mu(B(x,r)) = \mu((x-r, x+r) \cap [0,1]) = \mu((\max\{0, x-r\}, \min\{x+r, 1\})).$$

We consider four cases.

Case 1: $r \geq \max\{x, 1-x\} \implies r \geq \frac{1}{2}$
Thus $x - r \leq 0$ & $x + r \geq 1$ which shows that

$$\mu(B(x,r)) \leq \mu([0,1]) = 1 \leq 2 \cdot r^1.$$

Case 2: $r \leq \min\{x, 1-x\} \implies r \leq \frac{1}{2}$
Thus $1 \geq x - r \geq 0$ & $0 \leq x + r \leq 1$ which shows that

$$\mu(B(x,r)) = \mu((x-r, x+r)) = 2r^1.$$

Case 3: $x \leq r \leq 1 - x \implies x \leq \frac{1}{2}$
Since $r > 0$, $\exists \ n \in \mathbb{N}$ such that $r \geq \frac{1}{n}$. Thus $x - r \leq 0$ and $0 \leq x + r \leq 1$ which shows that

$$\mu(B(x,r)) = \mu([0, x+r)) = x + r \leq r + \frac{1}{2} \leq \left(\frac{n}{2} + 1\right) r^1.$$

Case 4: $x \geq r \geq 1 - x \implies x \geq \frac{1}{2}$
Since $r > 0$, $\exists \ n \in \mathbb{N}$ such that $r \geq \frac{1}{n}$. Thus $1 \geq x - r \geq 0$ and $x + r \geq 1$ which shows that

$$\mu(B(x,r)) = \mu((x-r, 1]) = 1 - x + r \leq r + \frac{1}{2} \leq \left(\frac{n}{2} + 1\right) r^1.$$

Thus our conditions for Frostman's Lemma are satisfied in all four cases which gives us $H^1(I) > 0$ which shows that $\dim_H(I) \geq 1$. □

We will now consider the box B in the next section.

3.2 Box $\mathbf{B} = [0,1]^2$

We will show that the topological, box and Hausdorff dimensions of B are all equal to 2. We formally state this in the main result below.

Theorem 3.2.1. *Let $B = [0,1]^2$. Then*

(a) $\dim_T(B) = 2$,

(b) $\dim_B(B) = 2$ *and*

(c) $\dim_H(B) = 2$.

3.2.1 Proof of Theorem 3.2.1(a)

We will show that $\dim_T(B) = 2$. Instead of constructing a refinement, we will use Lemma 2.0.30 in order to calculate the topological dimension of B. We know that $B = I \times I$ where $I = [0,1]$. Thus the topological dimension of B,

$$\dim_T(B) = \dim_T(I) + 1 = 1 + 1 = 2.$$

$\square$

3.2.2 Proof of Theorem 3.2.1(b)

In this section we will show that $\dim_B(B) = 2$. A good visual understanding is given in [FU16]. By Definition 2.0.32, the box dimension of B is given by

$$\lim_{\epsilon \downarrow 0} -\frac{\log N(\epsilon)}{\log \epsilon}$$

where $N(\epsilon)$ is the minimum number of boxes in $\mathbb{R}^2$ of size ϵ required to cover $[0,1]^2$.

Let $\epsilon > 0$ be given. A box of size ϵ in $\mathbb{R}^2$ must have side lengths of ϵ and $a \leq \epsilon$, the area of which equals $a \cdot \epsilon$ and is maximized at $a = \epsilon$. We claim that $\left\lceil \frac{1}{\epsilon^2} \right\rceil \leq N(\epsilon) \leq \left\lceil \frac{1}{\epsilon} \right\rceil^2$.

Suppose not and say there are $n < \left\lceil \frac{1}{\epsilon^2} \right\rceil$ box boxes of side ϵ that cover B. Note that the total area covered by all the boxes is $n \cdot \epsilon^2 < \frac{1}{\epsilon^2} \cdot \epsilon^2 = 1$. Thus they cannot cover $[0,1]^2$. This shows that the number of boxes required must be at least $\left\lceil \frac{1}{\epsilon^2} \right\rceil$. Now, we define

$$B_i := [i\epsilon, (i+1)\epsilon]$$

for all $0 \leq i \leq \left\lceil \frac{1}{\epsilon} \right\rceil - 1$. For any $(x,y) \in [0,1]^2$,

$$x \in \left[\left\lfloor \frac{x}{\epsilon} \right\rfloor \epsilon, \left(\left\lfloor \frac{x}{\epsilon} \right\rfloor + 1 \right) \epsilon \right] \text{ and } y \in \left[\left\lfloor \frac{y}{\epsilon} \right\rfloor \epsilon, \left(\left\lfloor \frac{y}{\epsilon} \right\rfloor + 1 \right) \epsilon \right].$$

This shows that

$$x \in B_{\lfloor \frac{x}{\epsilon} \rfloor} \text{ and } y \in B_{\lfloor \frac{y}{\epsilon} \rfloor} \implies (x,y) \in B_{\lfloor \frac{x}{\epsilon} \rfloor} \times B_{\lfloor \frac{y}{\epsilon} \rfloor}.$$

So

$$B \subset \left\{ B_i \times B_j \right\}_{0 \leq i,j \leq \left\lceil \frac{1}{\epsilon} \right\rceil - 1}.$$

Clearly the number of sets in this cover is $\left\lceil \frac{1}{\epsilon} \right\rceil \cdot \left\lceil \frac{1}{\epsilon} \right\rceil = \left\lceil \frac{1}{\epsilon} \right\rceil^2$. Thus we have $\left\lceil \frac{1}{\epsilon} \right\rceil^2$ boxes covering $[0,1]$ and so the claim that $\left\lceil \frac{1}{\epsilon^2} \right\rceil \leq N(\epsilon) \leq \left\lceil \frac{1}{\epsilon} \right\rceil^2$. holds. Note that $\left\lceil \frac{1}{\epsilon^2} \right\rceil = c \cdot \frac{1}{\epsilon^2}$ for some $c \geq 1$. Now

$$\frac{-\log \left\lceil \frac{1}{\epsilon^2} \right\rceil}{\log \epsilon} = \frac{\log c - 2\log \epsilon}{-\log \epsilon} = \frac{\log c}{-\log \epsilon} + 2.$$

As $\epsilon \downarrow 0$, we know that $\frac{\log c}{-\log \epsilon} \to 0$. Similarly, $\left\lceil \frac{1}{\epsilon} \right\rceil^2 = d \cdot \frac{1}{\epsilon^2}$ for some $d \geq 1$. Now

$$\frac{-\log \left\lceil \frac{1}{\epsilon} \right\rceil^2}{\log \epsilon} = \frac{\log d - 2\log \epsilon}{-\log \epsilon} = \frac{\log d}{-\log \epsilon} + 2.$$

As $\epsilon \downarrow 0$, we know that $\frac{\log d}{-\log \epsilon} \to 0$. Since both the bounds converge to 2 as $\epsilon \to 0$, by the application of squeeze theorem we can say that

$$\lim_{\epsilon \to 0} \frac{\log N(\epsilon)}{\log \frac{1}{\epsilon}}$$

exists and equals 2. This shows that $\dim_B(B)$ exists and is equal to 2.

□

3.2.3 Proof of Theorem 3.2.1(c)

In this section we shall show that B has a Hausdorff dimension of 2. By Definition 2.0.34, we need $d \geq 0$ such that the Hausdorff measure

$$H^s(B) = 0 \ \forall \ s > d \text{ and } H^s(B) = \infty \ \forall \ s > d.$$

We will first show that $\dim_H(B) \leq 2$.

Proof of $\dim_H(B) \leq 2$. This is equivalent to proving that $H^s(B) = 0 \ \forall \ s > 2$. Let $s > 2$ be arbitrary. Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of B, $\mathcal{U} = \{\mathcal{U}_i\}_{i \in I}$ satisfying $\operatorname{diam}(\mathcal{U}_i) \leq \delta \ \forall \ i \in I$, with I being a suitable index set. By Definition 2.0.34,

$$H^s(B) = \lim_{\delta \downarrow 0} \inf_{\mathcal{U} \in \mathcal{P}_\delta} \sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s$$

where $\sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n \to \infty} \sum_{\substack{i \in I_n \\ |I_n| < \infty \\ I_n \uparrow I}} (\operatorname{diam}(\mathcal{U}_i))^s$ if I is countably infinite.

Let $0 < \delta < 1$ be fixed. Define $\delta_0 = \frac{2}{\left\lceil \frac{2\sqrt{2}}{\delta} \right\rceil}$. Then $0 < \delta_0 \leq \frac{\delta}{\sqrt{2}}$ is such that $\frac{1}{\delta_0} = \frac{n}{2}$ for some $n \in \mathbb{N}$. We consider the open cover $\mathcal{B} = \left\{B_i \times B_j\right\}_{i,j \in J}$ of B such that

$$B_j = (i\delta_0, (i+1)\delta_0) \ \forall \ j \in J = \left\{-\frac{1}{2}, 0, \frac{1}{2}, \ldots, \frac{1}{\delta_0} - \frac{1}{2}, \frac{1}{\delta_0}\right\}.$$

Then the $\operatorname{diam}(B_i) = \sqrt{2}\delta_0 < \delta \ \forall \ i \in J$ and the minimum number of sets in $\mathcal{B}$ needed to cover B is $|J| = \left(\frac{2}{\delta_0} + 2\right)^2 = \left(\left\lceil \frac{2\sqrt{2}}{\delta} \right\rceil + 2\right)^2$. Since $\mathcal{B} \in \mathcal{P}_\delta$,

$$\inf_{\mathcal{U} \in \mathcal{P}_\delta} \sum_{i \in I} (\operatorname{diam}(\mathcal{U}_i))^s \leq \sum_{j \in J} (\operatorname{diam}(B_j))^s = |J| \cdot \delta^s \leq \left(\frac{2\sqrt{2}}{\delta} + 2\right)^2 \cdot \delta^s.$$

As $s > 2$, the right hand side of the above converges to zero as $\delta \downarrow 0$. From definition and an application of the squeeze theorem we have that

$$H^s(B) = 0.$$

Therefore $\dim_H(B) \leq 2$. □

We shall now show that $\dim_H(B) \geq 2$.

Proof of $\dim_H(B) \geq 2$. It is enough to prove that $H^2(B) > 0$. We use Frostman's Lemma in order to show this. We define a map $\mu : \mathcal{B}(B) \to [0,1]$ as the uniform distribution on $[0,1]^2$ given by:

$$\mu([a,b] \times [c,d]) = \int_c^d \int_a^b 1 \, dx dy \ = \ (d-c)(b-a) \ \ \forall \ 0 \leq a \leq b \leq 1 \text{ and } 0 \leq c \leq d \leq 1.$$

For $(x,y) \in B$ it is easy to deduce that

$$\mu(B((x,y),r)) = \mu(B((x,y),r) \bigcap [0,1]^2).$$

Then

$$\mu(B((x,y),r)) = \pi r^2 - \operatorname{Area}(B((x,y),r) \setminus [0,1]^2) \leq \pi r^2.$$

Thus our conditions for Frostman's Lemma are satisfied which gives us $H^2(B) > 0$ and we have thus shown that $\dim_H(B) \geq 2$. □

We have hence shown that $2 \leq \dim_H(B) \leq 2 \implies \dim_H(B) = 2$.

□

We have thus shown that the topological, box and Hausdorff dimensions of B are all equal to 2. The set that we will discuss in the next section is S^1.

3.3 Circle $\mathbf{S^1}$

In this section we shall show that the topological, box and Hausdorff dimensions of S^1 are all equal to 1. Let $S^1 \subset \mathbb{R}^2$ be given by $S^1 = \left\{x = (x_1, x_2) \in \mathbb{R}^2 : \|x\| = 1\right\}$, where $\|x\| := \sqrt{x_1^2 + x_2^2}$.

We define a map $f : [0, 2\pi) \to \mathbb{R}^2$ by

$$f(\theta) = (\cos\theta, \sin\theta). \tag{3.1}$$

It is easy to see that $\operatorname{Range}(f) = S^1$. Hence, S^1 can be viewed as a circle of radius 1 centered at the origin. Note that the curve f parameterizes the set S^1 with a single parameter θ, so one would guess that the set has dimension 1. Indeed, we will now prove that all the dimensions, namely the topological, box and Hausdorff dimensions of the circle are all equal to 1. We formally state this in the main result below.

Theorem 3.3.1. *Let* $S^1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$*. Then*

(a) $\dim_T(S^1) = 1$,

(b) $\dim_B(S^1) = 1$ *and*

(c) $\dim_H(S^1) = 1$.

3.3.1 Proof of Theorem 3.3.1(a)

We shall show that $\dim_T(S^1) = 1$. In order to show this we shall first show that $\dim_T(S^1) \geq 1$ and subsequently show that $\dim_T(S^1) \leq 1$.

Proof of $\dim_T(S) \geq 1$. We shall first show that S^1 is bounded, closed and path-connected in three steps.

Step 1 **Bounded:** By definition of S^1, $\forall\ x \in S^1$, $\|x\| = 1$. Therefore, S^1 is bounded trivially.

Step 2 **Closed:** Note that $(S^1)^c = \left\{x \in \mathbb{R}^2 : \|x\| < 1\right\} \bigcup \left\{x \in \mathbb{R}^2 : \|x\| > 1\right\}$. Now let $x_0 \in \left\{x \in \mathbb{R}^2 : \|x\| < 1\right\} \equiv B(0,1)$. We know that $B(0,1)$ is an open set by Lemma 2.0.3. So $\exists\ \epsilon > 0$ such that

$$B(x_0, \epsilon) \subset B(0,1) \subset S^1. \tag{3.2}$$

Let $x_0 \in \left\{x \in \mathbb{R}^2 : \|x\| > 1\right\}$. Then we may write $\|x_0\| = 1 + \epsilon$ for some $\epsilon > 0$. Consider $s \in B(x_0, \frac{\epsilon}{2})$. Observe by triangle inequality, we have

$$\|s\| \geq \|x_0\| - \|s - x_0\| \ = \ 1 + \epsilon - \frac{\epsilon}{2} > 1.$$

Therefore, $s \in \left\{x \in \mathbb{R}^2 : \|x\| > 1\right\}$ and consequently

$$B(x_0, \frac{\epsilon}{2}) \subset \left\{x \in \mathbb{R}^2 : \|x\| > 1\right\}. \tag{3.3}$$

Therefore for any arbitrary $x_0 \in (S^1)^c$, we have shown by (3.2) and (3.3) that $\exists\ \epsilon > 0$ such that $B(x_0, \frac{\epsilon}{2}) \subset (S^1)^c$. Hence $(S^1)^c$ is open which implies S^1 is closed.

Step 3 **Path-Connected:** Let $x, y \in S^1$. From (5) we know that $\exists\ \theta_1, \theta_2 \in [0, 2\pi)$ such that $x = (\cos(\theta_1), \sin(\theta_1))$ and $y = (\cos(\theta_2), \sin(\theta_2))$. We define $\gamma : [0,1] \to S^1$ by

$$\gamma(t) = (\cos(\theta_1 + t(\theta_2 + 2\pi - \theta_1)), \sin(\theta_1 + t(\theta_2 + 2\pi - \theta_1))).$$

Observe that $\gamma(0) = (\cos(\theta_1), \sin(\theta_1)) = x$ and $\gamma(1) = (\cos(2\pi + \theta_2), \sin(2\pi + \theta_2)) = y$. The 2π in definition of γ ensures that the path taken is counterclockwise. This is done so as to not create any path discontinuity from 2π to 0. Also since sin and cos are continuous functions, it is immediate that γ is continuous coordinate-wise which implies that γ is a continuous function. Therefore, x and y are connected via a continuous path γ in S^1. Since x and y were arbitrary, by definition, this proves that S^1 is path-connected.

As S^1 is a closed, bounded and path-connected (in particular connected) set of R^2, by Theorem 2.0.13 and Lemma 2.0.28, we have that $\dim_T(S^1) \geq 1$. $\square$

We shall now show the reverse inequality that $\dim_T(S^1) \leq 1$.

Proof of $\dim_T(S^1) \leq 1$. Let $\mathcal{U} = \{\mathcal{U}_i\}_{i \in I}$ be an open cover of S^1. Since S^1 is compact, $\mathcal{U}$ has a positive Lebesgue number $\delta > 0$ by Lemma 2.0.23. Let $0 < \delta_0 < \min\{\delta, 2\pi\}$ be such that $\frac{2\pi}{\delta_0} = \frac{n}{2}$ for some $n \in \mathbb{N}$. As stated before, from (3.1), we denote any element in S^1 as $x_\theta = (\cos\theta, \sin\theta)$ for some $\theta \in [0, 2\pi)$. We now define the set of sets $\{B_i\}_{i \in I}$ where

$$B_i = \left\{x_\theta \in S^1 \mid i\delta_0 < \theta < (i+1)\delta_0\right\} \ \forall\ i \in I = \left\{0, \frac{1}{2}, 1, \ldots, \frac{2\pi}{\delta_0}\right\}. \tag{3.4}$$

In order to show that $\dim_T(S^1) \leq 1$, we need the lemma below whose proof we will provide subsequently.

Lemma 3.3.2. *The following hold true for* $\{B_i\}_{i \in I}$*:*

(a) $diam(B_i) = \delta_0\ \forall\ i \in I$.

(b) B_i *is open in* $S^1\ \forall\ i \in I$.

(c) $\{B_i\}_{i \in I}$ *covers* S^1.

(d) $\{B_i\}_{i \in I}$ *has order 2.*

Assuming Lemma 3.3.2, we have constructed a refinement of an arbitrary open cover which has order 2. This implies that $\dim_T(S^1) \leq 1$. $\square$

As we have proved both inequalities, $\dim_T(S^1) = 1$.

Proof of Lemma 3.3.2. The proof of (a) is immediate as by definition, $\mathrm{diam}(B_i) = (i+1)\delta_0 - i\delta_0 = \delta_0$.

We now prove part (b). Let $x_\theta \in B_i$ for some $i \in I$. By definition of B_i, $i\delta_0 < \theta < (i+1)\delta_0$. Let $\theta = i\delta_0 + r$ for some $0 < r < \delta_0$. Then $\theta - i\delta_0 = r$ and $(i+1)\delta_0 - \theta = \delta_0 - r > 0$. Consider

$$\epsilon = \min\{r, \delta_0 - r\} > 0.$$

We have

$$i\delta_0 < \theta - \epsilon < \theta < \theta + \epsilon < (i+1)\delta$$

which implies that $\{x_\alpha \mid \alpha \in (\theta - \epsilon, \theta + \epsilon)\} \subset B_i$. Hence B_i is open and proof of part (a) is complete.

We will now prove part(c). Let $x_\theta \in S^1$ for some $\theta \in [0, 2\pi)$. It is then easy to see that $\theta \in \left(\left(\frac{1}{2}\left\lfloor\frac{2\theta}{\delta_0}\right\rfloor\right)\delta_0, \left(\frac{1}{2}\left\lfloor\frac{2\theta}{\delta_0}\right\rfloor + 1\right)\delta_0\right)$. By definition, $x_\theta \in B_{\frac{1}{2}\left\lfloor\frac{2\theta}{\delta_0}\right\rfloor}$. Therefore $S^1 \subset \bigcup_{i \in I} B_i$.

We now prove part (d). Let $\theta \in [0, 2\pi)$. If $\frac{\theta}{\delta_0} \in I$, then $x_\theta \in B_i$ with $i = \frac{\theta}{\delta_0} - \frac{1}{2}$. Otherwise if $\frac{\theta}{\delta_0} \notin I$, then $x_\theta \in B_i$ where $i = \left\lfloor\frac{2\theta}{\delta_0}\right\rfloor$ and $i = \left\lceil\frac{2\theta}{\delta_0} - 2\right\rceil$. Thus, each x_θ lies in at most two $B_i's$ which implies that the cover $\{B_i\}_{i \in I}$ has order 2. $\square$

$\square$

3.3.2 Proof of Theorem 3.3.1(b)

By Definition 2.0.32, the box dimension of S^1 is given by

$$-\lim_{\epsilon\downarrow 0}\frac{\log N(\epsilon)}{\log\epsilon} \tag{3.5}$$

where $N(\epsilon)$ is the minimum number of boxes (see Definition 2.0.31) of size $\epsilon > 0$ needed to cover S^1. So given an $\epsilon > 0$, we will try to cover S^1 with as few boxes as possible. It turns out that the best way to do this is to orient and place any box on S^1 so that the curve passes through the diagonal of the box (see Figure 3.2). The following lemma makes this observation precise.

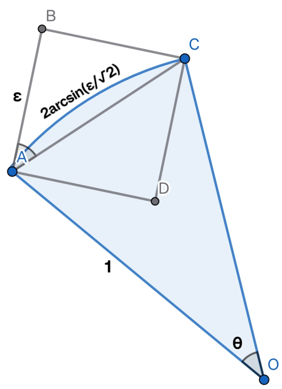


Figure 3.2: Arc Subtended by a Chord of Length $\sqrt{2}\epsilon$

Lemma 3.3.3. *Given $0 < \epsilon < \frac{\pi}{\sqrt{2}}$, a box of side ϵ intersecting S^1 at its diagonal vertices covers the maximum arc length of S^1 among all other boxes of size ϵ placed anywhere on S^1.*

Proof. Without loss of generality we may assume that a portion of the curve in S^1 intersects the box at two points. The diameter of a box in $\mathbb{R}^2$ of size ϵ is given by the length of its diagonal. Say the box passes through two points in S^1. Then the Euclidean distance between the points is the length of the chord of the arc passing through the box. The angle subtending a chord of length l on S^1 is equal to the arc length subtended by it and is given by $2\sin^{-1}\left(\frac{l}{2}\right)$, which is increasing in $0 < l < \pi$. In order to maximize the arc length covered by the box, we must maximize the chord length which therefore must be equal to the diameter of the box. Hence to maximise the arc length we shall place the box so that the curve in S^1 intersects the box at the diagonal vertices.

Given a box in $\mathbb{R}^2$ of size ϵ, its side lengths must be equal to a and ϵ where $0 < a \leq \epsilon$. Then its diameter is given by $\sqrt{a^2+\epsilon^2}$ which is clearly maximized at $a = \epsilon$. So, in order to maximise the length of curve inside, the box must be a box of length ϵ, the diameter of which is $\sqrt{2}\epsilon < \pi$. $\square$

Let $0 < \epsilon < \frac{\sqrt{2}}{\pi} < \frac{\pi}{\sqrt{2}}$ be given. Using the Lemma we shall use box boxes of side ϵ to cover S^1. From part of the proof of the lemma above, we know that the arc for the length covered is $2\sin^{-1}\left(\frac{\epsilon}{\sqrt{2}}\right)$. Therefore it is easy to see that

$$\exists\ c \geq 1 \text{ such that } N(\epsilon) \ = \ c \cdot \frac{2\pi}{2\sin^{-1}\left(\frac{\epsilon}{\sqrt{2}}\right)}.$$

Towards establishing (3.5), we consider

$$\frac{\log c + \log(2\pi) - \log\left(2\sin^{-1}\left(\frac{\epsilon}{\sqrt{2}}\right)\right)}{\log\epsilon}.$$

Note as $\epsilon \downarrow 0$, $\log\epsilon \to -\infty$ and therefore

$$\frac{\log c + \log(2\pi)}{\log\epsilon} \to 0. \tag{3.6}$$

As proved in [Hui], for $0 \leq x \leq 1$,

$$x \leq \sin^{-1}(x) \leq \frac{\pi}{2}x.$$

Thus as $0 < \epsilon < \frac{\sqrt{2}}{\pi}$,

$$\frac{\log\left(2\cdot\frac{\epsilon}{\sqrt{2}}\right)}{\log\epsilon} \leq \frac{\log\left(2\sin^{-1}\left(\frac{\epsilon}{\sqrt{2}}\right)\right)}{\log\epsilon} \leq \frac{\log\left(2\cdot\frac{\pi}{2}\cdot\frac{\epsilon}{\sqrt{2}}\right)}{\log\epsilon}.$$

Let $I = \left[0, \frac{\sqrt{2}}{\pi}\right)$. We define three functions f, g and h on $I \setminus \{0\}$ such that

$$f(x) = \frac{\log\left(\sqrt{2}\cdot x\right)}{\log(x)},\ g(x) = \frac{\log\left(2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\right)}{\log(x)} \text{ and } h(x) = \frac{\log\left(\frac{\pi}{\sqrt{2}}\cdot x\right)}{\log(x)}\ \forall\ x \in I \setminus \{0\}.$$

We know that 0 is a limit point of $I \setminus \{0\}$ as the sequence $\left\{\frac{1}{n+3}\right\}_{n\in\mathbb{N}}$ in $I \setminus \{0\}$ tends to 0. Let

$$L_1 := \lim_{\epsilon\downarrow 0} f(\epsilon) \text{ and } L_2 := \lim_{\epsilon\downarrow 0} h(\epsilon).$$

It is easy to see that L_1 and L_2 exist and are equal to 1. We will now apply the Squeeze Theorem in order to evaluate the box dimension of S^1.

Theorem 3.3.4. (Squeeze Theorem) *Let I be an interval containing $\{a\}$ and let f, g, h be intervals defined on $I \setminus \{a\}$. Say $\forall\ x \in I \setminus \{a\}$,*

$$f(x) \leq g(x) \leq h(x)$$

such that

$$\lim_{x\to a} f(x) = \lim_{y\to a} h(x) = L.$$

Then

$$\lim_{x\to a} g(x)$$

exists and is equal to L.

By the Squeeze Theorem,

$$\lim_{\epsilon\downarrow 0} g(\epsilon) = 1. \tag{3.7}$$

Placing (3.6) and (3.7) in (3.5), we have $\dim_B(S^1) = 1$.

3.3.3 Proof of Theorem 3.3.1(c)

By Definition 2.0.34, if d is the Hausdorff dimension of S^1, the Hausdorff Measure $H^s(S^1) = 0\ \forall\ s > d$ and $H^s(S^1) = \infty\ \forall\ s < d$.

Step I: We will prove that $\dim_H(S^1) \geq 1$.

Proof of $\dim_H(S^1) \geq 1$. We use Frostman's Lemma 2.0.35 for this part of the proof. We define a function $g : \mathbb{R} \to \mathbb{R}$ as the uniform distribution over $[0, 2\pi)$ given by:

$$g(\theta) = \begin{cases} \frac{1}{2\pi}, & \theta \in [0, 2\pi) \\ 0, & \text{else} \end{cases}$$

Then we define a Borel Measure (see Definition 2.0.33) $\mu : \mathcal{B}(S^1) \to [0,1]$ as:

$$\mu(\left\{x_\theta \in S^1 \mid 0 \leq a \leq \theta \leq b < 2\pi\right\}) = \int_a^b g(\theta) d\theta = \frac{b-a}{2\pi},$$

where $\mathcal{B}(S^1)$ is the set of all Borel subsets of S^1.

For $x_\theta \in S^1$ and $0 < r < 1$, it is easy to deduce using (3.1) that

$$B(x_\theta, r) = \left\{x_\alpha \mid 0 \leq \theta - 2\sin^{-1}\left(\frac{r}{2}\right) \leq \alpha \leq \theta + 2\sin^{-1}\left(\frac{r}{2}\right) < 2\pi\right\}.$$

Therefore,

$$\mu(B(x_\theta, r)) \ = 4\sin^{-1}\left(\frac{r}{2}\right) \cdot \frac{1}{2\pi} \ \leq \ 4 \cdot \frac{\pi}{2} \cdot \frac{r}{2} \cdot \frac{1}{2\pi} \ = \ \frac{1}{2} \cdot r^1,$$

where we have used the fact that $\frac{2}{\pi} x \leq \sin(x) \leq x\ \forall\ x \in\ [0, \frac{\pi}{2}]$.

And for $r > 1$, $\mu(B(x,r)) \leq 1 \leq r^1$ trivially. Therefore by the Frostman's lemma we can say that $\dim_H(S^1) \geq 1$. $\square$

Step II: We will now show that for $s > 1$, $H^s(S^1) = 0$.

Proof of $\dim_H(S^1) \leq 1$. This is equivalent to proving that $\dim_H(S^1) \leq 1$. Say $s = 1 + k$ for some $k > 0$. Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of S^1, $\mathcal{U} = \{\mathcal{U}_i\}_{i \in I}$ satisfying $\mathrm{diam}(\mathcal{U}_i) \leq \delta\ \forall\ i \in I$, with I being a suitable index set.

By Definition 2.0.34,

$$H^s(S^1) = \lim_{\delta \downarrow 0} \inf_{\mathcal{U} \in \mathcal{P}_\delta} \sum_{i \in I} (\mathrm{diam}(\mathcal{U}_i))^s$$

where $\sum_{i \in I} (\mathrm{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i \in I} (\mathrm{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n \to \infty} \sum_{\substack{i \in I_n \\ |I_n| < \infty \\ I_n \uparrow I}} (\mathrm{diam}(\mathcal{U}_i))^s$ if I is countably infinite. Let $\delta > 0$ be fixed. Consider an open cover $B = \left\{B_j\right\}_{j \in J}$ of S^1 such that

$$B_j = \left\{x_\theta \in S^1 \mid j\delta \leq \theta \leq (j+1)\delta\right\}\ \forall\ j \in J = \left\{0, \frac{1}{2}, 1, \ldots, \frac{\left\lceil \frac{4\pi}{\delta} \right\rceil}{2}\right\}.$$

Then the minimum number of sets in B needed to cover S^1 is $|J| \leq \frac{4\pi}{\delta} + 1$.

Since $B \in \mathcal{P}_\delta$,

$$\inf_{\mathcal{U} \in \mathcal{P}_\delta} \sum_{i \in I} (\mathrm{diam}(\mathcal{U}_i))^s \leq \sum_{j \in J} (\mathrm{diam}(B_j))^s = |J| \cdot \delta^s \leq 4\pi\delta^{s-1} + \delta^s.$$

As $s > 1$, the right hand side of the above converges to zero as $\delta \downarrow 0$. From definition and an application of the squeeze theorem we have that

$$H^s(S^1) = 0.$$

Therefore $\dim_H(S^1) \leq 1$. □

From Step 1 and Step 2 we have proved that $\dim_H(S^1) = 1$.

□

In this section, we used the definitions to calculate various dimensions of the non-fractal sets, I, B and S^1. Note that for these sets, the topological dimension was the same as the other dimensions. In the next section we shall see some sets for which this does not hold.

Chapter 4

Dimensions of C^α, S_3, S_4 and K

We shall define precisely and calculate the various dimensions of the fractals discussed in the introduction, namely the α–Cantor Set C^α, the Sierpiński Gasket S_3, the Sierpiński Carpet S_4 and the Koch Snowflake K.

4.1 The α-Cantor Set C^α

4.1.1 Formal Definition

For $\alpha \in (0,1)$, the α–Cantor Set is constructed recursively in the following way:

(i) C_0 is the interval [0,1].

(ii) We define the contraction maps

$$S_1 : [0,1] \to [0,1] \text{ as } S_1(x) = \frac{1-\alpha}{2} \cdot x \text{ and}$$

$$S_2 : [0,1] \to [0,1] \text{ as } S_2(x) = \frac{1-\alpha}{2} \cdot x + \frac{1+\alpha}{2}.$$

Then the n^{th} level Cantor Set

$$C_n := S_1(C_{n-1}) \cup S_2(C_{n-1}).$$

C^α is then defined as

$$C^\alpha := \bigcap_{n=0}^{\infty} C_n.$$

This shows that if

$$C_n \;=\; \bigsqcup_{i=1}^{2^i} [a_i, b_i],$$

we have

$$C_{n+1} = \bigsqcup_{i=1}^{2^i} \left(\left[a_i, a_i + (1-\alpha) \cdot \frac{b_i - a_i}{2} \right] \sqcup \left[a_i + (1+\alpha) \cdot \frac{b_i - a_i}{2}, b_i \right] \right). \tag{4.1}$$

4.1.2 Properties

We will show that C^α is an uncountable set of zero length.

Lemma 4.1.1. *Let C^α be the α–Cantor Set. Then C^α has zero length.*

Proof that C^α has zero length. Let C_n be constituted of N_n line segments of length l_n each. Then we have:

$$N_0 = 1 \text{ and } N_n = 2 \cdot N_{n-1} \ \forall\ n \geq 1.$$

Also,

$$l_0 = 1 \text{ and } l_n = \left(\frac{1-\alpha}{2}\right) l_n \ \forall\ n \geq 1.$$

The two recursions yield

$$N_n = 2^n \text{ and } l_n = \left(\frac{1-\alpha}{2}\right)^n \ \forall\ n \geq 0.$$

Therefore the perimeter of C_n equals $N_n \cdot l_n = (1-\alpha)^n \to 0$ as $n \to \infty\ \forall\ \alpha \in (0,1)$. Thus C^α has a total length of 0. □

Proof that C^α is uncountable. We construct a surjective map $f : C^\alpha \mapsto [0,1]^{\mathbb{N}}$. In order to do this we use Lemma 4.1.2 which we will assume for now and prove subsequently.

Lemma 4.1.2. *Let C^α be the Cantor Set. Then for all $n \geq 1$,*

$$C_n = \bigcup_{i_1, i_2, \ldots, i_n \in \{1,2\}} S_{i_n} \circ S_{i_{n-1}} \circ \ldots \circ S_{i_1}(C_0).$$

Let $x \in C^\alpha$. Then $x \in C_n\ \forall\ n \geq 0$. By Lemma 4.1.2, there exists a sequence $\{i_n\}_{n\in\mathbb{N}}$ such that

$$i_n \in \{1,2\}\ \forall\ n \in \mathbb{N} \text{ and } x = \lim_{n\to\infty} S_{i_n} \circ S_{i_{n-1}} \circ \ldots \circ S_{i_1}(y) \text{ for some } y \in [0,1].$$

We now define f as $f(x) = (i_1 - 1, i_2 - 1, \ldots)\ \forall\ x, y \in C^\alpha$. We will show that f is well-defined and surjective.

Well-Defined: Suppose $x \in C^\alpha$ has two sequences $\{i_n\}_{n\in\mathbb{N}}, \{j_n\}_{n\in\mathbb{N}}$ with

$$x \in I_n := S_{i_n} \circ \cdots \circ S_{i_1}(C_0) \text{ and } x \in J_n := S_{j_n} \circ \cdots \circ S_{j_1}(C_0),$$

but $\{i_n\}_{n\in\mathbb{N}} \neq \{j_n\}_{n\in\mathbb{N}}$. Let k be the smallest index such that $i_k \neq j_k$. We can assume without loss of generality that $i_k < j_k$. Then

$$I_k \cap J_k = \emptyset$$

because $S_{i_k} \circ \cdots \circ S_{i_1}(C_0) \subset \left[0, \frac{1-\alpha}{2}\right]$ and $S_{j_k} \circ \cdots \circ S_{j_1}(C_0) \subset \left[\frac{1+\alpha}{2}, 1\right]$. Hence $x \notin I_k \cap J_k$ which is a contradiction. Therefore, the sequence $\{i_n\}_{n\in\mathbb{N}}$ is unique for each $x \in C^\alpha$ and f is well-defined.

Surjective: Let $\{b_n\}_{n\in\mathbb{N}} \in \{0,1\}^{\mathbb{N}}$ and define $i_n := b_n + 1 \in \{1,2\}$. Let $y \in [0,1]$ be arbitrary and define

$$x_n := S_{i_n} \circ S_{i_{n-1}} \circ \cdots \circ S_{i_1}(y).$$

Since each S_i is a contraction with Lipschitz constant $r = \frac{1-\alpha}{2} < 1$, the composition

$$F_n := S_{i_n} \circ S_{i_{n-1}} \circ \cdots \circ S_{i_1}$$

is also a contraction with Lipschitz constant r^n. Then for any $m < n$,

$$|x_n - x_m| = |F_n(y) - F_m(y)| \le \operatorname{diam}(I_m) \le r^m \cdot \operatorname{diam}(C_0) = r^m.$$

Thus $\{x_n\}$ is a Cauchy sequence in $\mathbb{R}$ and hence converges. Thus the sequence $\{x_n\}$ converges to a point $x \in C^\alpha$. Let $x := \lim_{n\to\infty} x_n \in \mathbb{R}$ and

$$I_n := S_{i_n} \circ \cdots \circ S_{i_1}(C_0) \subset C_n \text{ by Lemma 4.1.2.}$$

We prove by induction that $x_n \in I_n \subset C_n$ for all $n \ge 1$.

Base case: $n = 1$. Then $x_1 = S_{i_1}(y) \in S_{i_1}(C_0) = I_1 \subset C_1$, since $y \in [0,1] = C_0$.

Inductive step: Assume $x_n \in I_n \subset C_n$. Then

$$x_{n+1} = S_{i_{n+1}}(x_n) \in S_{i_{n+1}}(I_n) = I_{n+1}.$$

Also, $I_{n+1} = S_{i_{n+1}}(I_n) \subseteq S_{i_{n+1}}(C_n) \subset C_{n+1}$. Hence $x_{n+1} \in I_{n+1} \subseteq C_{n+1}$. Thus by induction, $x_n \in C_n$ for all n, so $x = \lim_{n\to\infty} x_n \in \bigcap_{n=1}^\infty C_n = C^\alpha$. This shows that $f(x) = \{b_n\}_{n\in\mathbb{N}}$ and hence f is surjective.

Since f is well-defined and surjective, $|C^\alpha| \ge |\{0,1\}^{\mathbb{N}}|$ which is uncountable. Thus C^α is also uncountable. □

Therefore C^α is an uncountable set with 0 length which shows that its dimension must lie somewhere in $(0,1)$. The proof of Lemma 4.1.2 is provided below.

Proof of Lemma 4.1.2. We proceed by induction on n.

Base case: $n = 1$

$$C_1 = S_1(C_0) \cup S_2(C_0) = \bigcup_{i_1 \in \{1,2\}} S_{i_1}(C_0).$$

Inductive step: Assume the statement holds for some $n \ge 1$, i.e.,

$$C_n = \bigcup_{i_1,\ldots,i_n \in \{1,2\}} S_{i_n} \circ \cdots \circ S_{i_1}(C_0).$$

Then,

$$C_{n+1} = S_1(C_n) \cup S_2(C_n)$$

By the inductive hypothesis,

$$C_n = \bigcup_{i_1,\ldots,i_n \in \{1,2\}} S_{i_n} \circ \cdots \circ S_{i_1}(C_0).$$

So,

$$\begin{aligned}
C_{n+1} &= S_1\left(\bigcup_{i_1,\ldots,i_n} S_{i_n} \circ \cdots \circ S_{i_1}(C_0)\right) \cup S_2\left(\bigcup_{i_1,\ldots,i_n} S_{i_n} \circ \cdots \circ S_{i_1}(C_0)\right) \\
&= \bigcup_{i_{n+1} \in \{1,2\}} \bigcup_{i_1,\ldots,i_n \in \{1,2\}} S_{i_{n+1}} \circ S_{i_n} \circ \cdots \circ S_{i_1}(C_0) \\
&= \bigcup_{i_1,\ldots,i_{n+1} \in \{1,2\}} S_{i_{n+1}} \circ \cdots \circ S_{i_1}(C_0).
\end{aligned}$$

Thus, the result holds for $n+1$. By induction, the lemma is proved. □

Now we shall prove a crucial property of the $\frac{1}{3}$–Cantor Set.

Lemma 4.1.3. *Let C be the $\frac{1}{3}$–Cantor set defined by the contraction maps $\{S_i\}_{i=1}^2 : [0,1] \mapsto [0,1]$ given by*

$$S_1(x) = \frac{x}{3}\ ,\ S_2(x) = \frac{x}{3} + \frac{2}{3} \text{ and let}$$

$$C_0 := [0,1], \quad C_{n+1} := S_1(C_n) \cup S_2(C_n), \quad C := \bigcap_{n=0}^{\infty} C_n.$$

Then

$$C = \left\{ \sum_{k=1}^{\infty} \frac{a_k}{3^k} : a_k \in \{0,2\} \right\},$$

i.e., C consists of all points in $[0,1]$ whose ternary expansion contains no digit equal to 1.

Proof of Lemma 4.1.3. We proceed by induction to prove that C_n contains exactly those numbers in $[0,1]$ whose ternary expansion has no digit equal to 1 in the first n digits. We define the set

$$\mathcal{D}_n := \left\{ \sum_{k=1}^{n} \frac{a_k}{3^k} + \sum_{k=n+1}^{\infty} \frac{b_k}{3^k} : a_k \in \{0,2\},\ b_k \in \{0,1,2\} \right\}$$

and claim that

$$C_n = \mathcal{D}_n \cap [0,1].$$

Base case: $C_0 = [0,1] = \mathcal{D}_0 \cap [0,1]$ by definition of D_0.
Inductive step: Assume $C_n = \mathcal{D}_n \cap [0,1]$. Then

$$C_{n+1} = \frac{1}{3} C_n \cup \left(\frac{1}{3} C_n + \frac{2}{3} \right).$$

We compute:

$$\begin{aligned}
\frac{1}{3}\mathcal{D}_n &= \left\{ \sum_{k=1}^{n} \frac{a_k}{3^{k+1}} + \sum_{k=n+1}^{\infty} \frac{b_k}{3^{k+1}} : a_k \in \{0,2\},\ b_k \in \{0,1,2\} \right\} \\
&= \left\{ \sum_{k=1}^{n+1} \frac{a'_k}{3^k} + \sum_{k=n+2}^{\infty} \frac{b_k}{3^k} : a'_1 = 0,\ a'_k \in \{0,2\}\ (k>1) \right\} \text{ and} \\
\frac{1}{3}\mathcal{D}_n + \frac{2}{3} &= \left\{ \sum_{k=1}^{n} \frac{a_k}{3^{k+1}} + \sum_{k=n+1}^{\infty} \frac{b_k}{3^{k+1}} + \frac{2}{3} : a_k \in \{0,2\},\ b_k \in \{0,1,2\} \right\} \\
&= \left\{ \sum_{k=1}^{n+1} \frac{a'_k}{3^k} + \sum_{k=n+2}^{\infty} \frac{b_k}{3^k} : a'_1 = 2,\ a'_k \in \{0,2\}\ (k>1) \right\}.
\end{aligned}$$

Hence,

$$C_{n+1} = \mathcal{D}_{n+1} \cap [0,1].$$

Thus, by induction,

$$C_n = \left\{ x \in [0,1] : x = \sum_{k=1}^{\infty} \frac{a_k}{3^k},\ a_k \in \{0,2\}\ \forall\, 1 \le k \le n,\ a_j \in \{0,1,2\}\ \forall\, j > n \right\}.$$

Taking the intersection over all n, we obtain

$$C = \bigcap_{n=0}^{\infty} C_n = \left\{ \sum_{k=1}^{\infty} \frac{a_k}{3^k} : a_k \in \{0,2\} \right\}.$$

Thus we have shown that C consists of all points in $[0,1]$ whose ternary expansion can be represented by only using the digits 0 and 2. $\square$

In this section we shall prove the result stated below

Theorem 4.1.4. *Let C^α be the $\alpha-$Cantor Set. Then*

(a) $\dim_T(C^\alpha) = 0$,

(b) $\dim_B(C^\alpha) = \log_{\frac{2}{1-\alpha}}(2)$ *and*

(c) $\dim_H(C^\alpha) = \log_{\frac{2}{1-\alpha}}(2)$.

4.1.3 Proof of Theorem 4.1.4(a)

We shall prove that the α-Cantor Set $C^\alpha \subset \mathbb{R}$ is a compact and totally disconnected set in [0,1].

Step 1 **Bounded:** $C^\alpha \subset [0,1]$ which is a bounded set. Hence C^α is bounded.

Step 2 **Closed:** Notice that $C_1 \supsetneq C_2 \supsetneq C_3 \supsetneq \ldots$ where C_i is the set at the i^{th} iteration. Therefore,

$$C^\alpha = \bigcap_{i\geq 1} C_i \implies (C^\alpha)^c = \bigcup_{i\geq 1} C_i^c$$

Now, C_i is a finite union of closed intervals by [4.1] which is closed $\forall\ i \implies C_i^c$ is open $\forall\ i \implies (C^\alpha)^c$ is the arbitrary union of open sets, which is open. Hence C^α is closed.

Step 3 **Totally Disconnected:** Let $y > x \in C^\alpha$. By our construction in (4.1), we know that the length of each interval in C_i is $l_i = \left(\frac{1-\alpha}{2}\right)^i$. Since $\frac{1-\alpha}{2} < 1$, we have $l_i \to 0$ as $i \to \infty$. Hence $\forall\ \epsilon > 0$, $\exists\ N \in \mathbb{N}$ such that $\forall\ i \geq N$, $l_i < \epsilon$. Also, the distance between two consecutive intervals of C_i can be calculated to be $d_i := \alpha \cdot \left(\frac{1-\alpha}{2}\right)^{i-1}$.

Consider $0 < \epsilon < y - x$. Then there will exist i such that $l_i < \epsilon < y - x$ which means that there will be disjoint intervals $I_1, I_2 \subset C_i$ with $x \in I_1$ and $y \in I_2$. If $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ where $a_1 < b_1 < b_1 + d_i \leq a_2 < b_2$, we define

$$A := \left[0, b_1 + \frac{d_i}{2}\right) \text{ and } B := \left(b_1 + \frac{d_i}{2}, 1\right],$$

which are relatively open in $[0,1]$. Clearly, $I_1 \subset A$, $I_2 \subset B$ and

$$A \sqcup B = [0,1] \setminus \left\{ b_i + \frac{d_i}{2} \right\} \implies (A \cap C^\alpha) \bigsqcup (B \cap C^\alpha) = C^\alpha$$

since $b_i + \frac{d_i}{2} \notin C_i \supset C^\alpha$.

Hence, we get a separation between x and y for all $x \neq y \in C^\alpha$ which shows that C^α is totally disconnected.

Since $C^\alpha \subset \mathbb{R}$ is closed, bounded and totally disconnected, it has topological dimension 0 by Theorem 2.0.13 and Lemma 2.0.28.

$\square$

4.1.4 Proof of Theorem 4.1.4(b)

By definition, the box dimension of C^α is given by

$$-\lim_{\epsilon\to 0}\frac{\log(N(\epsilon))}{\log(\epsilon)}$$

where $N(\epsilon)$ is the minimum number of boxes of size $\epsilon > 0$ needed to cover C^α. We will estimate $N_n(\epsilon)$, the minimum number of boxes of size ϵ required to cover C_n, the n^{th} polygonal approximation of C^α. Since we are concerned about the limit as $\epsilon \to 0$, we can assume $\epsilon \leq 2^{-n}$, given n. This ensures that the size of the box is lesser than each individual interval. By a method similar to Lemma 3.5, it can be shown that boxes of side ϵ placed such that the intervals pass through the diagonals minimizes $N_n(\epsilon)$. At the n^{th} iteration, we have 2^n line segments of length $(\frac{1-\alpha}{2})^n$.

By lengthwise comparison,

$$N_n(\epsilon) = t\cdot\frac{2^n\cdot(\frac{1-\alpha}{2})^n}{\sqrt{2}\epsilon} = t\cdot\frac{(1-\alpha)^n}{\sqrt{2}\epsilon} \text{ for some } t \geq 1.$$

Then

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = -\frac{\log(t)+n\log(1-\alpha)-\log(\sqrt{2}\epsilon)}{\log(\sqrt{2}\epsilon)}.$$

Let $\epsilon = c\cdot 2^{-n}$ for some $0 < c \leq 1$. Then

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = -\frac{\log(t)+n\log(1-\alpha)-\log(c\sqrt{2}\cdot 2^{-n})}{\log(c\sqrt{2}\cdot 2^{-n})}.$$

It is easy to see that as $n \to \infty$, the above limit exists and is equal to $\log_{\frac{2}{1-\alpha}}(2)$. Thus $\dim_B(C^\alpha) = \log_{\frac{2}{1-\alpha}}(2)$.

$\square$

4.1.5 Proof of Theorem 4.1.4(c)

By Definition 2.0.34, we need $d \ni H^s(C^\alpha) = 0\ \forall\ s > d$ and $d \ni H^s(C^\alpha) = \infty\ \forall\ s < d$.

We claim that $\dim_H(C^\alpha) = \log_{(\frac{2}{1-\alpha})}(2)$. In order to show that we will show first that $\dim_H(C^\alpha) \leq \log_{(\frac{2}{1-\alpha})}(2)$ and subsequently show that $\dim_H(C^\alpha) \geq \log_{(\frac{2}{1-\alpha})}(2)$.

Proof of $\dim_H(C^\alpha) \leq \log_{(\frac{2}{1-\alpha})}(2)$. It is enough to show that for all $s > \log_{(\frac{2}{1-\alpha})}(2)$, $H^s(C^\alpha) = 0$. Say $s = \log_{(\frac{2}{1-\alpha})}(2) + k$ for some $k > 0$.

Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of C^α, $\mathcal{U} = \{\mathcal{U}_i\}_{i\in I}$ satisfying $\operatorname{diam}(\mathcal{U}_i) \leq \delta\ \forall\ i \in I$, with I being a suitable index set.

By Definition 2.0.34,

$$H^s(C^\alpha) = \lim_{\delta\downarrow 0}\inf_{\mathcal{U}\in\mathcal{P}_\delta}\sum_{i\in I}(\operatorname{diam}(\mathcal{U}_i))^s$$

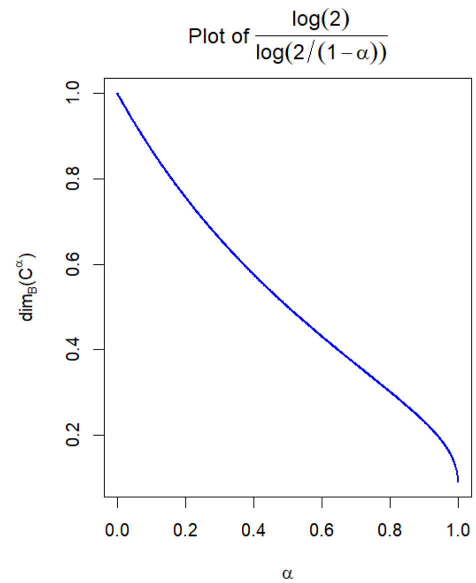


Figure 4.1: Plot of α vs $\dim_B(C^\alpha)$

where $\sum_{i\in I}(\text{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i\in I}(\text{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n\to\infty}\sum_{\substack{i\in I_n\\|I_n|<\infty\\I_n\uparrow I}}(\text{diam}(\mathcal{U}_i))^s$ if I is countably infinite. Let $\delta>0$ be fixed. Then

$$\inf_{\mathcal{U}\in\mathcal{P}_\delta}\sum_i(\text{diam}(\mathcal{U}_i))^s \;\le\; \sum_{i=1}^{N(\delta)}\delta^s$$

where $N(\delta)$ is the minimum number of open sets of diameter δ required to cover C^α. In order to calculate $N(\delta)$, we need the minimum number of open sets of diameter δ, $N_n(\delta)$, that can cover C_n. By lengthwise comparison, $N_n(\delta)=t\cdot\frac{(1-\alpha)^n}{\delta}$ for some $t>0$.

Let $n\in\mathbb{N}$ be fixed. Since we are concerned with $\delta\;\to\;0$, we choose $\delta>0$ such that

$$\delta=c\cdot\left(\frac{1-\alpha}{2}\right)^{-n}\text{ for some }c\in(0,1].$$

Then

$$H^s(C^\alpha)\;\le\;\lim_{n\to\infty}\;t\cdot(1-\alpha)^n\delta^{(s-1)}\;\ni\;\delta=c\cdot\left(\frac{1-\alpha}{2}\right)^{-n}.$$

The limit above tends to 0 which shows that $H^s(C^\alpha)\;=\;0.$ □

Proof of $\dim_H(C^\alpha)\ge\log_{\left(\frac{2}{1-\alpha}\right)}(2)$. We use Frostman's Lemma for this part of the proof. We define $\mu_n:\mathcal{B}(C_n)\to[0,1]$ as a uniform mass distribution over C_n such that $\forall\; i$, $\mu_n(I_i)\;=\;2^{-n}$ where I_i is the i^{th} interval of C_n. Then μ_n can be extended to $\mu:\mathcal{B}(C^\alpha)\to[0,1]$ by

$$\mu(E)\;=\;\lim_{n\to\infty}\mu_n(E\cap C_n)\;\forall\;E\subset\mathcal{B}(C^\alpha).$$

Then, $\forall\ r < 1$, we can choose n such that

$$\left(\frac{1-\alpha}{2}\right)^n \leq r < \left(\frac{1-\alpha}{2}\right)^{n-1}.$$

Let l_n be the length of each interval in C_n. By lengthwise comparison, $B(x,r)$ contains at most $\frac{2r}{l_n}$ segments of C_n. Note that $\frac{2r}{l_n} < \frac{4}{1-\alpha}$ using the upper bound of r and the fact that $l_n = \left(\frac{1-\alpha}{2}\right)^n$. Thus

$$\mu(B(x,r)) \ \leq \ \frac{4}{(1-\alpha)2^n} \ \leq \left(\frac{4}{1-\alpha}\right)\cdot r^{\log_{\left(\frac{2}{1-\alpha}\right)}(2)}.$$

And when $r \geq 1$, $\mu(B(x,r)) \ = 1 \ \leq \ r^{\log_{\left(\frac{2}{1-\alpha}\right)}(2)}$. Therefore by the Frostman's lemma we can say that $\dim_H(C^\alpha) \geq \log_{\left(\frac{2}{1-\alpha}\right)}(2)$. □

Hence, $\dim_H(C^\alpha) \ = \ \log_{\left(\frac{2}{1-\alpha}\right)}(2)$.

□

4.2 The Sierpiński Gasket S_3

4.2.1 Formal Definition

The Sierpiński Gasket is constructed as follows:

Let Δ be an equilateral triangle with vertices $P = \begin{pmatrix}0\\0\end{pmatrix}$, $Q = \begin{pmatrix}1\\0\end{pmatrix}$ and $R = \begin{pmatrix}\frac{1}{2}\\\frac{\sqrt{3}}{2}\end{pmatrix}$.

Let T_0 be the region enclosing and including the triangle Δ.

Consider $f_1, f_2, f_3 : T_0 \mapsto \mathbb{R}^2$ given by

$$f_1\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{2}\begin{pmatrix}x\\y\end{pmatrix}, \quad \forall \begin{pmatrix}x\\y\end{pmatrix} \in T_0,$$

$$f_2\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{2}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}\frac{1}{2}\\0\end{pmatrix}, \quad \forall \begin{pmatrix}x\\y\end{pmatrix} \in T_0,$$

$$f_3\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{2}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}\frac{1}{4}\\\frac{\sqrt{3}}{4}\end{pmatrix}, \quad \forall \begin{pmatrix}x\\y\end{pmatrix} \in T_0.$$

Now we define T_n recursively as

$$T_n = f_1(T_{n-1}) \cup f_2(T_{n-1}) \cup f_3(T_{n-1}).$$

Thus we have $T_0 \supset T_1 \supset T_2 \supset \ldots$ and we can define

$$S_3 = \bigcap_{n=0}^{\infty} T_n.$$

4.2.2 Properties

We shall show that S_3 is a set of infinite perimeter and zero area.

Lemma 4.2.1. *Let S_3 be the Sierpiński Gasket. Then*

(a) S_3 has infinite perimeter.

(b) S_3 has zero area.

Proof that S_3 has infinite perimeter. Let the perimeter of T_n be denoted by P_n. We know that T_0 is an equilateral triangle of side 1 and thus $P_0 = 3$. We claim that for all $n > 0$,

$$P_n = 3 \cdot \left(1 + \frac{1}{2}\sum_{k=0}^{n-1}\left(\frac{3}{2}\right)^k\right).$$

Base Case ($n = 1$): T_1 is the triangle T_0 with the central inverted equilateral removed. Thus two boundaries, the boundary of T_0 and the boundary of the removed triangle both contribute to the perimeter of T_1. The boundary of the removed triangle is an equilateral triangle of side $\frac{1}{2}$ and thus has perimeter $\frac{3}{2}$.

$$P_1 = 3 + \frac{3}{2} = 3 \cdot \left(1 + \frac{1}{2}\left(\frac{3}{2}\right)^0\right).$$

Inductive Step: Assume it holds for $n = m$, i.e.,

$$P_m = 3 \cdot \left(1 + \frac{1}{2}\sum_{k=0}^{m-1}\left(\frac{3}{2}\right)^k\right).$$

Then at step $m+1$, we divide each of the 3^m triangles of T_m into 4 further sub-triangles and remove the central sub-triangle from each triangle. Thus we remove 3^m sub-triangles of side length $\frac{1}{2^{m+1}}$. The additional contribution to the perimeter is:

$$P_{m+1} = P_m + 3 \cdot \frac{1}{2} \cdot \left(\frac{3}{2}\right)^m = 3 \cdot \left(1 + \frac{1}{2}\sum_{k=0}^{m}\left(\frac{3}{2}\right)^k\right).$$

So by induction, the formula holds. Since $\frac{3}{2} > 1$, the geometric sum diverges:

$$\sum_{k=0}^{n-1}\left(\frac{3}{2}\right)^k \to \infty \quad \text{as } n \to \infty.$$

Hence,

$$P_n \to \infty.$$

□

Proof that S_3 has zero area. Let T_n be the union of N_n equilateral triangles each of side length l_n. Then we have:

$$N_0 = 1 \text{ and } N_n = 3 \cdot N_{n-1} \ \forall \ n \geq 1.$$

Also,

$$l_0 = 1 \text{ and } l_n = \frac{l_{n-1}}{2} \ \forall\ n \geq 1.$$

Also, two recursions yield

$$N_n = 3^n \text{ and } l_n = 2^{-n} \ \forall\ n \geq 0.$$

The area of each triangle in T_n is given by $\frac{\sqrt{3}}{4 \cdot 3^2 \cdot 2^{2n}}$. Thus the total area of triangles of T_n equals $N_n \cdot \left(\frac{\sqrt{3}}{4} l_n^2\right) = \frac{\sqrt{3}}{4}\left(\frac{3}{4}\right)^n$ which converges to 0 as $n \to \infty$. Thus S_3 has zero area. $\square$

The two results above shows that S_3 must have a dimension in $(1, 2)$. In this section we shall prove the result stated below

Theorem 4.2.2. *Let S_3 be the Sierpiński Gasket. Then*

(a) $\dim_T(S_3) = 1$,

(b) $\dim_B(S_3) = \log_2(3)$ *and*

(c) $\dim_H(S_3) = \log_2(3)$.

4.2.3 Proof of Theorem 4.2.2(a)

We will show that S_3 has a topological dimension of 1. In order to show this we will first show that $\dim_T(S_3) \geq 1$ and subsequently show that $\dim_T(S_3) \leq 1$.

Proof of $\dim_T(S_3) \geq 1$.. We will prove that $S_3 \subset \mathbb{R}^2$ is a closed, bounded and connected set.

Step 1 **Bounded:** Let the circumcentre of T_0 be defined as the origin. The circumradius of T_0 is $r := \frac{1}{3\sqrt{3}}$. By the construction, $T_0 \supset T_1 \supset T_2 \supset \ldots$. Hence, S_3 is bounded by any ball containing the circumcircle of T_0, i.e., $S_3 \subset B(0, r + \epsilon)\ \forall\ \epsilon > 0$. Hence, S_3 is bounded.

Step 2 **Closed:** We know that $S_3 = \bigcap_{n=0}^{\infty} T_n$. Then $S_3^c = \bigcup_{n=0}^{\infty} T_n^c$.

(i) Base Case: For $n = 0$, T_0 is closed as it is defined to enclose and contain Δ.

(ii) Let T_n be closed for some n. Then $T_{n+1} = f_1(T_n) \cup f_2(T_n) \cup f_3(T_n)$. It is easy to see that for all $1 \leq i \leq 3$, the restricted maps $f_i : T_n \to f_i(T_n)$ are continuous bijective functions with continuous inverses and hence map closed sets to closed sets by definition of continuity. Thus T_{n+1} is the finite union of closed sets, which is closed.

$\therefore$ S_3 is the arbitrary intersection of closed sets, which is closed.

Step 3 **Connected:** We assume for contradiction that S_3 is disconnected. Then there exist relatively open subsets $A, B \subset S_3$ such that:

(a) $S_3 = A \cup B$.

(b) $A \cap B = \emptyset$.

(c) $A \neq \emptyset$ and $B \neq \emptyset$.

By definition of relative openness, there exist open sets $U, V \subset \mathbb{R}^2$ such that:

$$A = S_3 \cap U \quad \text{and} \quad B = S_3 \cap V.$$

Since S_3 is compact (by Theorem 2.0.13) and A, B are closed in S_3 (since their complements B and A are relatively open in S_3), they are compact. By Lemma 2.0.14:

$$\delta := d(A, B) = \inf\{\|a - b\| : a \in A, b \in B\} > 0.$$

We choose n large enough so that the side length 2^{-n} of triangles in T_n satisfies $2^{-n} < \delta/2$.

For any triangle Δ in T_n intersecting S_3:

- $\operatorname{diam}(\Delta \cap S_3) = 2^{-n} < \delta/2$.
- If $\Delta \cap S_3 \cap A \neq \emptyset$ and $\Delta \cap S_3 \cap B \neq \emptyset$, then $\exists\ a \in A$, $b \in B$ with $d(a, b) \leq \operatorname{diam}(\Delta) < \delta$, contradicting $d(A, B) = \delta$.

Thus each $\Delta \cap S_3$ lies entirely in A or B.

This would require T_n to be disconnected if both A and B are non-empty. However, we shall prove that T_n is connected for all $n \in \mathbb{N}$, which will lead to a contradiction.

(i) Base Case: $n = 0$. T_0 which is a solid, closed, equilateral triangle. Let $x, y \in T_0$ and define $\gamma : [0, 1] \to T_0$ by $\gamma(t) = (1 - t)x + ty$. Notice that γ is continuous and that $\gamma(0) = x$ and $\gamma(1) = y$. Hence, it is a continuous path from x to y. Since x, y were arbitrary, T_0 is path-connected, which implies it is connected.

(ii) Inductive Case: Let T_n be connected for some n. Then by construction, T_{n+1} is the formed by joining three scaled copies of T_n via the maps f_1, f_2 and f_3. The coordinates of the vertices of T_n are given by $(x_1, y_1) = P$, $(x_2, y_2) = Q$ and $(x_3, y_3) = R$. Then the coordinates of its 3 subtriangles in T_{n+1} are given by:

$f_1(T_n)$: Centered at (x_1, y_1), with vertices:

$$(x_1, y_1), \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right), \left(\frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2}\right).$$

$f_2(T_n)$: Centered at (x_2, y_2), with vertices:

$$(x_2, y_2), \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right), \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}\right).$$

$f_3(T_n)$: Centered at (x_3, y_3), with vertices:

$$(x_3, y_3), \left(\frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2}\right), \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}\right).$$

We can see that no sub-triangle is disjoint from the union of the other two as there is a vertex in common. Then $T_{n+1} = T_1 \cup T_2 \cup T_3$, each of which is a scaled copy of T_n, which is connected by hypothesis. Thus T_1, T_2 and T_3 are also connected.

Hence, $T_{n+1} \subset \mathbb{R}^2$ satisfies the conditions for Lemma 2.0.20, implying it is connected. The assumption therefore leads to a contradiction, so S_3 must be connected.

We have proved above that S_3 is a closed, bounded and connected subset of $\mathbb{R}^2$, which by Lemma 2.0.28 proves that $\dim_T(S_3) \geq 1$. $\square$

Now we will prove the reverse inequality that $\dim_T(S_3) \leq 1$.

Proof of $\dim_T(S_3) \leq 1$. We know that S_3 is a compact subset of $\mathbb{R}^2$ which means that all open covers of S_3 have a positive Lebesgue Number. Let $\mathcal{U}$ be an arbitrary cover of S_3 and $\delta > 0$ be its Lebesgue number. We choose $n \ni d_n := \frac{1}{3 \cdot 2^n} < \delta$.

Then we work with the triangle at the n^{th} iterative step T_n, where we have $N_n = 3^n$ triangles of diameter $d_n < \delta$. Let $\{x_1, x_2, \ldots, x_{N_n}\}$ be the circumcentres of the triangles $\{t_1, t_2, \ldots, t_{N_n}\}$ of T_n. It is easy to show that T_n has $M_n = \frac{3(3^{n+1}-1)}{2}$ vertices. Let us name the vertices $\{v_1, v_2, \ldots, v_{M_n}\}$.

Let $1 \leq i \leq N_n$ be fixed. Then the open ball $B_i := B\left(x_i, \frac{d_n}{\sqrt{3}}\right)$ forms the circumcircle of t_i. Let t_i have vertices $v_{i_1}, v_{i_2}, v_{i_3}$ where $1 \leq i_1, i_2, i_3 \leq M_n$. Then $B_i \bigcup \left\{B\left(v_{i_j}, \frac{d_n}{2\sqrt{3}}\right)\right\}_{j=1}^{3}$ covers t_i.

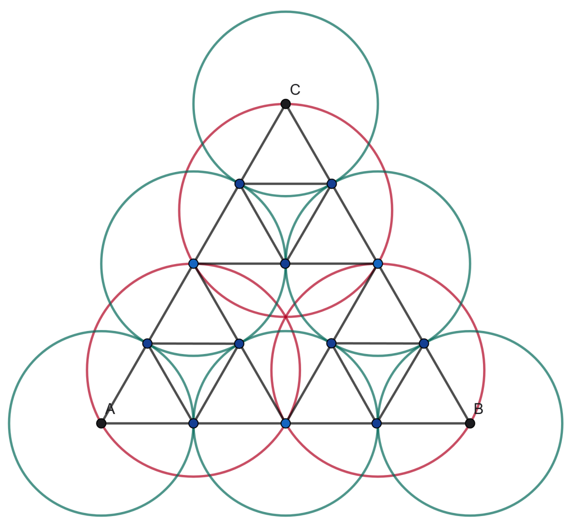


Figure 4.2: Refinement of a Cover of T_1

For all $1 \leq j \leq M_n$, we define the set $C_j := B\left(v_j, \frac{d_n}{2\sqrt{3}}\right)$. We define

$$\mathcal{V} := \{B_i \cap S_3\}_{i=1}^{N_n} \bigcup \{C_j \cap S_3\}_{j=1}^{M_n}$$

(see Figure 4.2 where the red balls are the $B_i's$ and the green balls are the $C_j's$) which covers S_3 by construction. By the definition of Lebesgue number, all elements of $\mathcal{V}$ are contained in some set of $\mathcal{U}$. For all $1 \leq i \leq N_n$, all non-vertex points of t_i are contained in B_i uniquely and in at most one set of $\{C_j\}_{j=1}^{M_n}$ since the $C_j's$ are pairwise-disjoint. For all $1 \leq j \leq M_n$, the vertex $v_j \in C_j$. Hence given any arbitrary cover of S_3, we have refined it to a cover of order 2 and by definition, $\dim_T(S_3) \leq 1$. $\square$

The two proofs above tell us that $\dim_T(S_3) = 1$.

$\square$

Below, we will calculate the two fractal dimensions, namely the box and Hausdorff dimensions and show that they are all equal to $\log_2(3) \in (1, 2)$.

4.2.4 Proof of Theorem 4.2.2(b)

By Definition 2.0.32, the box dimension of S_3 is given by

$$-\lim_{\epsilon \to 0} \frac{\log(N(\epsilon))}{\log(\epsilon)}$$

where $N(\epsilon)$ is the minimum number of boxes of size $\epsilon > 0$ needed to cover S_3. We will estimate $N_n(\epsilon)$, the minimum number of boxes of size ϵ required to cover T_n, the n^{th} polygonal approximation of S_3. Since we are concerned about the limit as $\epsilon \to 0$, we can assume $\epsilon \leq 2^{-n}$, given n. At iteration n, we have 3^n equilateral triangles, each of length $\frac{1}{3 \cdot 2^n}$. Hence, the area of each triangle =

$$\frac{\sqrt{3}}{4 \cdot 3^2 \cdot 2^{2n}}.$$

Comparing area wise,

$$N_n(\epsilon) = t \cdot \frac{3^n \cdot \frac{\sqrt{3}}{4 \cdot 3^2 \cdot 2^{2n}}}{\epsilon^2}$$

for some $t \geq 1$ where $\frac{\sqrt{3}}{4 \cdot 3^2 \cdot 2^{2n}}$ is the area of T_n and ϵ^2 is the maximum area of a box of size ϵ. Then we evaluate

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = \frac{\log(t) + \log\left(\frac{\sqrt{3}}{36}\right) + n\log(3) - 2n\log(2) - 2\log(\epsilon)}{\log(\epsilon)}.$$

We can assume $\epsilon = c \cdot 2^{-n}$ for some $0 < c \leq 1$. Then

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = -\frac{\log(N_n(c \cdot 2^{-n}))}{\log(c \cdot 2^{-n})} = \frac{\log(t) + \log\left(\frac{\sqrt{3}}{36}\right) + n\log 3 - 2n\log(2) - 2\log(c) + 2n\log(2)}{\log(c) - n\log(2)}.$$

By simple computation, it is easy to see that as $n \to \infty$, the limit above exists and is equal to $\dim_B(S_3) = \log_2(3) \approx 1.5849\ldots$

$\square$

4.2.5 Proof of Theorem 4.2.2(c)

By Definition 2.0.34, we need $d \ni H^s(S_3) = 0 \ \forall\ s > d$ and $d \ni H^s(S_3) = \infty \ \forall\ s < d$.

We claim that $\dim_H(S_3) = \log_2(3)$. In order to show that we will show first that $\dim_H(S_3) \geq \log_2(3)$ and subsequently show that $\dim_H(S_3) \leq \log_2(3)$.

Proof of $\dim_H(S_3) \geq \log_2(3)$. We use Frostman's Lemma for this part of the proof. We define $\mu_n : \mathcal{B}(T_n)$ as a uniform mass distribution over T_n such that

$$\mu(t_i) \ = \ 3^{-n} \ \forall \ i,$$

where t_i is the i^{th} triangle of T_n. Then μ_n can be extended to $\mu : \mathcal{B}(S_3) \to [0,1]$ by

$$\mu(E) \ = \ \lim_{n\to\infty} \mu_n(E \cap T_n) \ \forall \ E \in \mathcal{B}(S_3).$$

Then given $r < 1$, we can choose n such that

$$\frac{2^{-n}}{6} \leq r < \frac{2^{-n+1}}{6}.$$

The area of $B(x,r)$ is $\pi \cdot r^2$ whereas the area of a triangle in T_n is given by $\frac{\sqrt{3}}{36\cdot 2^{2n}}$. Thus $B(x,r)$ contains at most $\frac{\pi\cdot r^2}{\left(\frac{\sqrt{3}}{36\cdot 2^{2n}}\right)}$ triangles of T_n. Using the upper bound of r, $\frac{\pi\cdot r^2}{\left(\frac{\sqrt{3}}{36\cdot 2^{2n}}\right)} < \frac{4\pi}{\sqrt{3}}$. Then

$$\mu(B(x,r)) \ \leq \ \frac{4\pi}{\sqrt{3}} \cdot 3^{-n} \ = \ 4\sqrt{3}\pi \cdot 3^{\log_2(3)} \cdot \left(\frac{2^{-n}}{6}\right)^{\log_2(3)} \ \leq \ 4\sqrt{3}\pi \cdot 3^{\log_2(3)} \cdot r^{\log_2(3)}.$$

And when $r > 1$, we have

$$\mu(B,r) \ \leq \ 1 \ \leq \ r^{\log_2(3)}.$$

Therefore by the Frostman's lemma we can say that $\dim_H(S_3) \ \geq \ \log_2(3)$. □

Proof of $\dim_H(S_3) \leq \log_2(3)$. It is enough to show that for all $s > \log_2(3)$, $H^s(S_3) = 0$. Say $s = \log_2(3) + k$ for some $k > 0$.

Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of S_3, $\mathcal{U} \ = \ \{\mathcal{U}_i\}_{i\in I}$ satisfying $\mathrm{diam}(\mathcal{U}_i) \leq \delta \ \forall \ i \in I$, with I being a suitable index set.

By Definition 2.0.34,

$$H^s(S_3) = \lim_{\delta\downarrow 0} \inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_{i\in I} (\mathrm{diam}(\mathcal{U}_i))^s$$

where $\sum_{i\in I}(\mathrm{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i\in I}(\mathrm{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n\to\infty} \sum_{\substack{i\in I_n \\ |I_n|<\infty \\ I_n\uparrow I}} (\mathrm{diam}(\mathcal{U}_i))^s$ if I is countably infinite. Let $\delta > 0$ be fixed. Then

$$\inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_i (\mathrm{diam}(\mathcal{U}_i))^s \ \leq \ \sum_{i=1}^{N(\delta)} \delta^s$$

where $N(\delta)$ is the minimum number of open sets of diameter δ required to cover S_3. In order to calculate $N(\delta)$, we need the minimum number of open sets of diameter δ, $N_n(\delta)$, that can cover T_n. Comparing the areas of a triangle and the diameter of a ball of radius $\frac{\delta}{2}$,

$$N_n(\delta) = t \cdot \frac{\left(3^n \cdot \frac{\sqrt{3}}{4^n}\right)}{\left(\frac{\pi\delta^2}{4}\right)} \text{ for some } t > 1.$$

Let $n \in \mathbb{N}$ be fixed. Since we are concerned with $\delta \to 0$, we choose $\delta > 0$ such that

$$\delta = c \cdot 2^{-n} \text{ for some } c \in (0,1].$$

Then

$$H^s(S_3) \leq \lim_{n\to\infty} \frac{4\sqrt{3}}{\pi} \cdot t \cdot \left(\frac{3}{4}\right)^n \delta^{s-2} \ni \delta = c \cdot 2^{-n}.$$

The limit above tends to 0 which shows that $H^s(S_3) = 0$. □

Thus we have shown that $\dim_H(S_3) = \log_2(3)$.

□

We have hence proved Theorem 4.2.2.

□

4.3 The Sierpiński Carpet S_4

4.3.1 Formal Definition

$U_0 = [0,1]\times[0,1] \subset \mathbb{R}^2$ be the unit box. Define 8 contractive affine maps $\{f_i\}_{i=1}^8 : S \to S$ by:

$$\begin{aligned}
f_1(x,y) &= \left(\frac{x}{3}, \frac{y}{3}\right), & f_2(x,y) &= \left(\frac{x}{3}+\frac{1}{3}, \frac{y}{3}\right),\\
f_3(x,y) &= \left(\frac{x}{3}+\frac{2}{3}, \frac{y}{3}\right), & f_4(x,y) &= \left(\frac{x}{3}, \frac{y}{3}+\frac{1}{3}\right),\\
f_5(x,y) &= \left(\frac{x}{3}+\frac{2}{3}, \frac{y}{3}+\frac{1}{3}\right), & f_6(x,y) &= \left(\frac{x}{3}, \frac{y}{3}+\frac{2}{3}\right),\\
f_7(x,y) &= \left(\frac{x}{3}+\frac{1}{3}, \frac{y}{3}+\frac{2}{3}\right), & f_8(x,y) &= \left(\frac{x}{3}+\frac{2}{3}, \frac{y}{3}+\frac{2}{3}\right).
\end{aligned}$$

Then the Sierpiński carpet at the n^{th} level

$$U_n := \bigcup_{i=1}^{8} f_i(U_{n-1})$$

and the Sierpiński carpet is defined as

$$S_4 := \bigcap_{n=0}^{\infty} U_n.$$

4.3.2 Properties

We will prove that S_4 is a set of infinite perimeter and zero area.

Lemma 4.3.1. *Let S_4 be the Sierpiński Carpet. Then*

(a) S_4 has infinite perimeter.

(b) S_4 has zero area.

Proof that S_4 has infinite perimeter. Let the perimeter of U_n be denoted by P_n. We know that U_0 is the box $[0,1]^2$ and thus $P_0 = 4$. We claim that for all $n > 0$,

$$P_n = 4 \cdot \left(1 + \frac{1}{3}\sum_{k=0}^{n-1}\left(\frac{8}{3}\right)^k\right).$$

We prove this by induction.

Base Case ($n = 1$): U_1 is the box U_0 with the central sub-box removed. Thus two boundaries, the boundary of U_0 and the boundary of the removed box both contribute to the perimeter of U_1. The boundary of the removed box is a box of side $\frac{1}{3}$ and thus has perimeter $\frac{4}{3}$.

$$P_1 = 4 + \frac{4}{3} = 4 \cdot \left(1 + \frac{1}{3} \cdot \left(\frac{8}{3}\right)^0\right).$$

Inductive Step: Assume it holds for $n = m$, i.e.,

$$P_m = 4 \cdot \left(1 + \frac{1}{3}\sum_{k=0}^{m-1}\left(\frac{8}{3}\right)^k\right).$$

Then at step $m+1$, we divide each of the 8^m boxes of U_m into 9 further sub-boxes and remove the central sub-box from each box. Thus we remove 8^m sub-boxes of side length $\frac{1}{3^{m+1}}$. The additional contribution to the perimeter is:

$$\Delta P = 4 \cdot \left(\frac{1}{3}\left(\frac{8}{3}\right)^m\right).$$

Thus

$$P_{m+1} = P_m + \Delta P = 4 \cdot \left(1 + \frac{1}{3}\sum_{k=0}^{m}\left(\frac{8}{3}\right)^k\right).$$

Hence, the formula holds by induction. Observe that $\frac{8}{3} > 1$, so the sum diverges:

$$\sum_{k=0}^{n-1}\left(\frac{8}{3}\right)^k \to \infty \quad \text{as } n \to \infty.$$

Therefore,

$$P_n \to \infty.$$

□

Proof that S_4 has zero area. Let U_n be the union of N_n boxes each of side length l_n. Then we have:

$$N_0 = 1 \text{ and } N_n = 8 \cdot N_{n-1} \ \forall\ n \geq 1.$$

Also,

$$l_0 = 1 \text{ and } l_n = \frac{l_{n-1}}{3} \ \forall\ n \geq 1.$$

Also, two recursions yield

$$N_n = 8^n \text{ and } l_n = 3^{-n} \ \forall\ n \geq 0.$$

The area of each box of U_n is given by 3^{-2n}. Thus the total area of triangles of U_n equals $N_n \cdot l_n^2 = \left(\frac{8}{9}\right)^n$ which converges to 0 as $n \to \infty$. Thus S_4 has zero area. □

The two results above shows that S_4 must have a dimension in $(1,2)$. Now we will show an interesting characteristic of S_4.

Lemma 4.3.2. *Let $S_4 \subset [0,1]^2$ be the Sierpiński carpet defined by the contraction maps $\{S_{i,j}\}_{(i,j)\in I}$ with index set*

$$I = \{(i,j) \in \{0,1,2\}^2 : (i,j) \neq (1,1)\},$$

and each map $S_{i,j}(x,y) = \left(\frac{x+i}{3}, \frac{y+j}{3}\right)$. Define

$$U_0 := [0,1]^2, \quad U_{n+1} := \bigcup_{(i,j)\in I} S_{i,j}(U_n), \quad S_4 := \bigcap_{n=0}^{\infty} U_n.$$

Then

$$S_4 = \left\{ \sum_{k=1}^{\infty} \left(\frac{a_k}{3^k}, \frac{b_k}{3^k} \right) : (a_k, b_k) \in I \ \forall k \in \mathbb{N} \right\},$$

i.e., S_4 consists of all points in $[0,1]^2$ whose base-3 coordinates avoid the pair $(1,1)$ at every digit.

Proof of Lemma 4.3.2. We define the set

$$\mathcal{D}_n := \left\{ \sum_{k=1}^{n} \left(\frac{a_k}{3^k}, \frac{b_k}{3^k} \right) + \sum_{k=n+1}^{\infty} \left(\frac{u_k}{3^k}, \frac{v_k}{3^k} \right) : (a_k, b_k) \in I,\ (u_k, v_k) \in \{0,1,2\}^2 \right\}$$

and claim that

$$U_n = \mathcal{D}_n \cap [0,1]^2.$$

Base case: $U_0 = [0,1]^2 = \mathcal{D}_0 \cap [0,1]^2$ by definition.
Inductive step: Assume $U_n = \mathcal{D}_n \cap [0,1]^2$. Then

$$U_{n+1} = \bigcup_{(i,j)\in I} \left(\frac{1}{3} U_n + \left(\frac{i}{3}, \frac{j}{3} \right) \right).$$

Now,

$$\begin{aligned}
\frac{1}{3}\mathcal{D}_n &= \left\{ \sum_{k=1}^{n} \left(\frac{a_k}{3^{k+1}}, \frac{b_k}{3^{k+1}} \right) + \sum_{k=n+1}^{\infty} \left(\frac{u_k}{3^{k+1}}, \frac{v_k}{3^{k+1}} \right) : (a_k, b_k) \in I,\ (u_k, v_k) \in \{0,1,2\}^2 \right\} \\
&= \left\{ \sum_{k=1}^{n+1} \left(\frac{a'_k}{3^k}, \frac{b'_k}{3^k} \right) + \sum_{k=n+2}^{\infty} \left(\frac{u_k}{3^k}, \frac{v_k}{3^k} \right) : (a'_k, b'_k) \in I \text{ for } 1 \le k \le n+1,\ (u_k, v_k) \in \{0,1,2\}^2 \right\}.
\end{aligned}$$

Adding $\left(\frac{i}{3}, \frac{j}{3}\right)$ to all elements restricts the new digit to $(a'_1, b'_1) = (i,j) \in I$ and all further $(a'_k, b'_k) \in I$. Thus

$$U_{n+1} = \mathcal{D}_{n+1} \cap [0,1]^2.$$

By induction,

$$U_n = \left\{ (x,y) \in [0,1]^2 : (x,y) = \sum_{k=1}^{\infty} \left(\frac{a_k}{3^k}, \frac{b_k}{3^k} \right),\ (a_k, b_k) \in I \ \forall\, 1 \le k \le n,\ (u_k, v_k) \in \{0,1,2\}^2 \ \forall\, k > n \right\}.$$

Taking the intersection over all n, we obtain

$$S_4 = \bigcap_{n=0}^{\infty} U_n = \left\{ \sum_{k=1}^{\infty} \left(\frac{a_k}{3^k}, \frac{b_k}{3^k} \right) : (a_k, b_k) \in I \right\}.$$

$\square$

Thus S_4 contains points $(x,y) \in [0,1]^2$ with the property that on expanding $x = 0.a_1 a_2 \ldots$ and $y = 0.b_1 b_2 \ldots$ in base 3, $(a_k, b_k) \neq (1,1)\ \forall\ k \in \mathbb{N}$. In this section we will prove the result stated below.

Theorem 4.3.3. *Let S_4 be the Sierpiński Carpet. Then*

(a) $\dim_T(S_4) = 1$,

(b) $\dim_B(S_4) = \log_3(8)$ *and*

(c) $\dim_H(S_4) = \log_3(8)$.

4.3.3 Proof of Theorem 4.3.3(a)

We will start by proving that $\dim_T(S_4) \geq 1$ and then subsequently prove that $\dim_T(S_4) \leq 1$.

Proof of $\dim_T(S_4) \geq 1$. Let $I = \{(i,j) \in \{0,1,2\}^2 : (i,j) \neq (1,1)\}$ and let the contraction maps $S_{i,j}(x,y) = \left(\frac{x+i}{3}, \frac{y+j}{3} \right)$. Define the iterations:

$$U_0 := [0,1]^2, \quad U_{n+1} := \bigcup_{(i,j) \in I} S_{i,j}(U_n), \quad S_4 := \bigcap_{n=0}^{\infty} U_n.$$

We proceed by induction to show:

$$\{0\} \times [0,1] \subset U_n \quad \text{for all } n \in \mathbb{N}.$$

Base case: $U_0 = [0,1]^2 \supset \{0\} \times [0,1]$.
Inductive step: Suppose $\{0\} \times [0, \frac{1}{3}] \subset U_n$. Then note that

$$\{0\} \times [0,1] \subset U_n,$$

since $\{0\} \times [0, \frac{1}{3}] \subset U_n$ and $U_n \supset U_0 = [0,1]^2$. Now apply the maps:

$$S_{0,0}(\{0\} \times [0,1]) = \left\{ \left(\tfrac{0}{3}, \tfrac{y+0}{3} \right) : y \in [0,1] \right\} = \{0\} \times \left[0, \tfrac{1}{3} \right],$$

$$S_{0,1}(\{0\} \times [0,1]) = \{0\} \times \left[\tfrac{1}{3}, \tfrac{2}{3} \right], \quad S_{0,2}(\{0\} \times [0,1]) = \{0\} \times \left[\tfrac{2}{3}, 1 \right].$$

Thus,

$$\{0\} \times [0,1] \subset S_{0,0}(U_n) \cup S_{0,1}(U_n) \cup S_{0,2}(U_n) \subset U_{n+1}.$$

Hence, by induction, for all n, $\{0\} \times [0,1] \subset U_n$, so

$$\{0\} \times [0,1] \subset \bigcap_{n=0}^{\infty} U_n = S_4.$$

We know that $[0,1]$ and $\{0\}\times[0,1]$ are homeomorphic sets by the map $f:[0,1]\mapsto\{0\}\times[0,1]$ given by $f(x)=(0,x)$. By Lemma 2.0.26,

$$\dim_T(\{0\}\times[0,1]) = \dim_T([0,1]) = 1.$$

By Lemma 2.0.25,

$$S_4 \supset \{0\}\times[0,1] \implies \dim_T(S_4) \geq \dim_T(0\times[0,1]) = 1$$

and we are done. □

Proof of $\dim_T(S_4) \leq 1$. We begin by showing that S_4 is a compact subset of $\mathbb{R}^2$. By Theorem 2.0.13, it is enough to show that S_4 is closed and bounded.

(a) **Bounded:** By construction, S_4 is contained in the unit box $[0,1]^2 \subset \mathbb{R}^2$, hence bounded.

(b) **Closed:** S_4 is defined as the intersection of a nested sequence of closed subsets of $[0,1]^2$, namely

$$S_4 = \bigcap_{n=0}^{\infty} U_n,$$

where U_n is the n^{th} approximation obtained by recursively removing the open central box from each box in the previous stage. Each U_n is a finite union of closed boxes, hence closed. Since countable intersections of closed sets are closed in $\mathbb{R}^n$, S_4 is closed.

Thus S_4 is compact. By Lemma 2.0.23, every open cover of S_4 admits a positive Lebesgue number. Let $\mathcal{U}$ be an arbitrary open cover of S_4, and let $\delta > 0$ be the Lebesgue number for $\mathcal{U}$. That is, for every $x \in S_4$, the open ball $B(x,\delta) \subseteq U$ for some $U \in \mathcal{U}$.
We choose $n \in \mathbb{N}$ such that

$$\sqrt{2}\cdot 3^{-n+1} < \delta.$$

Let $U_n \subset [0,1]^2$ be the level-n approximation of the Sierpiński carpet: the union of 8^n retained boxes of side length 3^{-n} (marked black in Figure 4.3). At each stage, the central open box is removed from each subdivided 3×3 block. Thus, the number of removed open boxes is

$$R := \frac{8^n - 1}{7}.$$

Let $x_1, x_2, \ldots, x_R \in \mathbb{R}^2$ denote the centers of the removed boxes. We join these centres to form a grid of $(\sqrt{R}-1)^2$ open boxes (marked red in Figure 4.3). Let $\mathcal{R}$ denote the collection of the interiors of these boxes. The elements of $\mathcal{R}$ are pairwise disjoint and cover the central region of S_4.

To cover the remaining portion of S_4 outside the red grid, we define a collection $\mathcal{G}$ of open sets (marked green in Figure 4.3). The goal is to complete the red grid of $(\sqrt{R}-1)^2$ open boxes to one of $(\sqrt{R}+1)^2$ open boxes that will cover all points of U_n except the edges in between.

Let $x^{(1)}_{\min}$ and $x^{(1)}_{\max}$ denote the minimum and maximum x-coordinates among the centers $\{x_1, x_2, \ldots, x_R\}$, and similarly let $x^{(2)}_{\min}$ and $x^{(2)}_{\max}$ denote the corresponding y-coordinates. The red grid lies inside the open box

$$(x^{(1)}_{\min}, x^{(1)}_{\max}) \times (x^{(2)}_{\min}, x^{(2)}_{\max}).$$

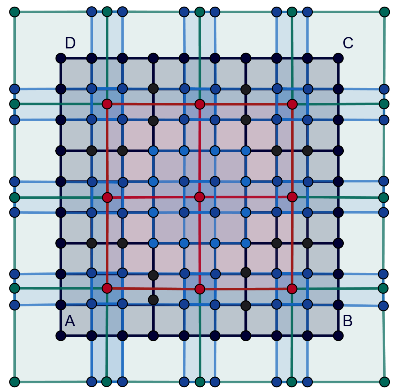


Figure 4.3: Refinement of a Cover of U_2

We fix $\epsilon = \frac{3^{-n+1}}{2}$ and then tile each of the four surrounding strips

$$\begin{aligned}
G_1 &:= (0-\epsilon, 1+\epsilon) \times (0-\epsilon, x_{\min}^{(2)}), && \text{(bottom strip)}\\
G_2 &:= (0-\epsilon, 1+\epsilon) \times (x_{\max}^{(2)}, 1+\epsilon), && \text{(top strip)}\\
G_3 &:= (0-\epsilon, x_{\min}^{(1)}) \times (x_{\min}^{(2)}, x_{\max}^{(2)}), && \text{(left strip)}\\
G_4 &:= (x_{\max}^{(1)}, 1+\epsilon) \times (x_{\min}^{(2)}, x_{\max}^{(2)}), && \text{(right strip).}
\end{aligned}$$

into smaller open green boxes, each congruent to a red box. Let $\mathcal{G}$ denote the collection of all such boxes. Then:

- each element of $\mathcal{G}$ is open and pairwise disjoint,
- $\mathcal{R} \cup \mathcal{G}$ forms a full $(\sqrt{R}+1)^2$ grid of open boxes with uniform mesh,
- the union of all boxes in $\mathcal{G}$ covers all of S_4 outside the red grid, except for the edges between them, which are handled separately.

Now, the interiors of red and green sets cover all of S_4 except points lying on the edges of the red and green boxes. To complete the cover, define $\mathcal{B}$ (marked by blue in Figure 4.3). For each horizontal edge e in the grid formed by centers x_i, x_j (where x_i and x_j lie on the same horizontal line), let the corners of the corresponding removed boxes r_i and r_j be denoted

$$c_{i,1}, c_{i,2}, c_{i,3}, c_{i,4} \quad \text{and} \quad c_{j,1}, c_{j,2}, c_{j,3}, c_{j,4}$$

indexed counterclockwise from the bottom-left corner. Define

$$B_{ij} := (c^{(x)}_{i,1}, c^{(x)}_{j,2}) \times (c^{(y)}_{i,1}, c^{(y)}_{i,4}),$$

an open box along the horizontal edge between them. Define vertical strips analogously.

Let $\mathcal{B}$ denote the collection of all such horizontal and vertical edge-covering open boxes.

Claim 1. *$\mathcal{R} \cup \mathcal{G} \cup \mathcal{B}$ is an open cover of S_4.*

Proof. Every point in S_4 either:

- lies inside a red box (i.e., in the interior of a grid cell bounded by the centres of removed boxes),
- lies inside a green box outside the red grid, or
- lies along an edge of some red or green box which is covered by some blue box in $\mathcal{B}$.

Hence the union of the sets in $\mathcal{R} \cup \mathcal{G} \cup \mathcal{B}$ covers S_4. □

Claim 2. *The cover has order at most 2.*

Proof. All the boxes in the union of the red and green sets ($\mathcal{R} \cup \mathcal{G}$) are constructed to be pairwise disjoint. The blue boxes are also pairwise disjoint in S_4. Hence, any point in S_4 lies in at most (one red xor green set) or one blue set. Thus, the cover has order at most 2. □

Claim 3. *The diameter of each set in the cover is strictly less than δ.*

Proof. The diameter of each red and green box is $\sqrt{2} \cdot 3^{-n+1} < \delta$ and the diameter of each blue box is $\sqrt{(3^{-n})^2 + (4 \cdot 3^{-n})^2} = \sqrt{17} \cdot 3^{-n} < \delta$. □

Thus, $\mathcal{R} \cup \mathcal{G} \cup \mathcal{B}$ is an open refinement of $\mathcal{U}$ of order at most 2 and mesh $< \delta$. Hence, by definition of topological dimension,

$$\dim_T(S_4) \leq 1.$$

□

This proves Theorem 4.3.3(a). □

Below, we will calculate the two fractal dimensions, namely the box and Hausdorff dimensions and show that they are all equal to $\log_3(8) \in (1, 2)$.

4.3.4 Proof of Theorem 4.3.3(b)

By Definition 2.0.32, the box dimension of S_4 is given by

$$-\lim_{\epsilon \to 0} \frac{\log(N(\epsilon))}{\log(\epsilon)}$$

where $N(\epsilon)$ is the minimum number of boxes of size $\epsilon > 0$ needed to cover S_4. We will estimate $N_n(\epsilon) :=$ The minimum number of boxes of side ϵ required to cover U_n, the n^{th} polygonal approximation of S_4. Since we are concerned about the limit as $\epsilon \to 0$, we can assume $\epsilon \leq 3^{-n}$, given n. Then, the boxes may be placed such that the boxes completely cover the carpet boxes, in order to minimize $N_n(\epsilon)$.

Then

$$N_n(\epsilon) = t \cdot \frac{8^n \cdot \frac{1}{9^n}}{\epsilon^2} \text{ for some } t \geq 1,$$

where $(\frac{8}{9})^n$ is the area of U_n & ϵ^2 the maximum area of a box of size ϵ. Then we compute

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = \frac{\log(t) + n\log\left(\frac{8}{9}\right) - \log(\epsilon)}{\log(\epsilon)}.$$

We assume $\epsilon = c \cdot 3^{-n}$ for some $0 < c \leq 1$. Thus

$$-\frac{\log(t) + n\log\left(\frac{8}{9}\right) - \log(\epsilon)}{\log(\epsilon)} = -\frac{\log(t) + n\log\left(\frac{8}{9}\right) - \log(c) + n\log(3)}{n\log(3) - \log(c)}.$$

It is easy to see that as $n \to \infty$, the above limit exists and is equal to

$$\dim_B(S_4) \ = \ \log_3(8) \ \approx \ 1.8927\ldots$$

$\square$

4.3.5 Proof of Theorem 4.3.3(c)

By Definition 2.0.34, we need $d \ni H^s(S_4) = 0 \ \forall \ s > d$ and $d \ni H^s(S_4) = \infty \ \forall \ s < d$. We claim that $\dim_H(S_4) \ = \ \log_3(8)$. In order to show that we will show first that $\dim_H(S_4) \geq \log_3(8)$ and subsequently show that $\dim_H(S_4) \leq \log_3(8)$.

Proof of $\dim_H(S_4) \geq \log_3(8)$. We use Frostman's Lemma for this part of the proof. We define $\mu_n : \mathcal{B}(U_n)$ as a uniform mass distribution over U_n such that

$$\mu(s_i) \ = \ 8^{-n} \ \forall \ i,$$

where s_i is the i^{th} box of U_n. Then μ_n can be extended to $\mu : \mathcal{B}(S_4) \to [0,1]$ by

$$\mu(E) \ = \ \lim_{n\to\infty} \mu_n(E \cap U_n) \ \forall \ E \in \mathcal{B}(S_4).$$

Then given $r < 1$, we can choose n such that

$$3^{-n} \leq r < 3^{-n+1}.$$

The area of $B(x,r)$ is $\pi \cdot r^2$ whereas the area of a box in U_n is given by $\left(\frac{8}{9}\right)^n$. Thus $B(x,r)$ contains at most $\frac{\pi \cdot r^2}{\left(\frac{8}{9}\right)^n}$ boxes of U_n. Using the upper bound of r, $\frac{\pi \cdot r^2}{\left(\frac{8}{9}\right)^n} \leq 9\pi$. Then

$$\mu(B(x,r)) \ \leq \ 9\pi \cdot 8^{-n} \ = \ 9\pi \cdot (3^{-n})^{\log_3(8)} \ \leq \ 9\pi \cdot r^{\log_3(8)}.$$

And when $r > 1$, we have

$$\mu(B,r) \ \leq \ 1 \ \leq \ r^{\log_3(8)}.$$

Therefore by the Frostman's lemma we can say that $\dim_H(S_4) \ \geq \ \log_3(8)$. $\square$

Proof of $\dim_H(S_4) \le \log_3(8)$. It is enough to show that for all $s > \log_3(8)$, $H^s(S_4) = 0$. Say $s = \log_3(8) + k$ for some $k > 0$.

Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of S_4, $\mathcal{U} = \{\mathcal{U}_i\}_{i\in I}$ satisfying $\mathrm{diam}(\mathcal{U}_i) \le \delta\ \forall\ i \in I$, with I being a suitable index set.

By Definition 2.0.34,

$$H^s(S_4) = \lim_{\delta\downarrow 0} \inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_{i\in I} (\mathrm{diam}(\mathcal{U}_i))^s$$

where $\sum_{i\in I}(\mathrm{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i\in I}(\mathrm{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n\to\infty} \sum_{\substack{i\in I_n \\ |I_n|<\infty \\ I_n\uparrow I}} (\mathrm{diam}(\mathcal{U}_i))^s$ if I is countably infinite. Let $\delta > 0$ be fixed. Then

$$\inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_i (\mathrm{diam}(\mathcal{U}_i))^s \le \sum_{i=1}^{N(\delta)} \delta^s$$

where $N(\delta)$ is the minimum number of open sets of diameter δ required to cover S_4. In order to calculate $N(\delta)$, we need the minimum number of open sets of diameter δ, $N_n(\delta)$, that can cover U_n. Comparing the areas of a box in U_n and the diameter of a ball of radius $\frac{\delta}{2}$,

$$N_n(\delta) = t \cdot \frac{\left(\frac{8}{9}\right)^n}{\left(\frac{\pi\delta^2}{4}\right)} \text{ for some } t > 0.$$

Let $n \in \mathbb{N}$ be fixed. Since we are concerned with $\delta \to 0$, we choose $\delta > 0$ such that

$$\delta = c \cdot 3^{-n} \text{ for some } c \in (0,1].$$

Then

$$H^s(S_4) \le \lim_{n\to\infty} \delta^{s-2} \cdot t \cdot \left(\frac{8}{9}\right)^n \frac{4}{\pi}.$$

As $n \to \infty$, the limit above tends to 0 which shows that $H^s(S_4) = 0$. $\square$

Thus we have shown that $\dim_H(S_4) = \log_3(8)$.

$\square$

We have hence proved Theorem 4.3.3.

$\square$

4.4 The Koch Snowflake K

4.4.1 Formal Definition

We define the following lines L_1, L_2 and L_3 as:

$$L_1 = \mathcal{L}((0,0),(1,0)),\ L_2 = \mathcal{L}\left((0,0),\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)\right) \text{ and } L_3 = \mathcal{L}\left(\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right),(1,0)\right).$$

Let

$$\mathbf{S} = \left\{S_j : \mathbb{R}^2 \to \mathbb{R}^2\right\}_{j=1}^4 \text{ such that } S_j(x^T) = \frac{1}{3}O_j(x^T) + b_j^T$$

where

$$b_1 = (0,0),\ b_2 = \left(\frac{1}{3}, 0\right),\ b_3 = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right),\ b_4 = \left(\frac{2}{3}, 0\right)$$

and

$$O_1 = O_4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ O_2 = \begin{bmatrix} \cos\frac{\pi}{3} & \sin\frac{\pi}{3} \\ -\sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{bmatrix},\ O_3 = \begin{bmatrix} \cos\frac{-\pi}{3} & \sin\frac{-\pi}{3} \\ -\sin\frac{-\pi}{3} & \cos\frac{-\pi}{3} \end{bmatrix}.$$

Fix $1 \le i \le 3$ and let $K_i'^{(1)} = L_i$. We define the sequence of sets $\left\{K_i'^{(n)}\right\}_{n\ge1}$ as

$$K_i'^{(n)} = \mathbf{S}\left(K_i'^{(n-1)}\right) = \bigcup_{j=1}^{4} S_j\left(K_i'^{(n-1)}\right)\ \forall\ n > 1.$$

Then the Koch Curve K_i generated by L_i is defined as

$$K_i' = \lim_{n\to\infty} K_i'^{(n)} = \lim_{n\to\infty} \mathbf{S}^n\left(K_i'^{(1)}\right) = \bigcap_{m=1}^{\infty} \bigcup_{n\ge m} K_i'^{(n)}.$$

The Koch Snowflake $K := K_1' \cup K_2' \cup K_3'$, i.e., it is created by joining the three rotated and reflected copies of K'. We define the n^{th} polygonal approximation of K to be

$$K_n := K_1'^{(n)} \cup K_2'^{(n)} \cup K_3'^{(n)}.$$

4.4.2 Properties

We will show that K is a set with infinite perimeter and zero area.

Lemma 4.4.1. *Let K be the Koch Snowflake. Then*

(a) K has infinite perimeter.

(b) K has zero area.

Proof that K has infinite perimeter. Let K_n be the union of N_n line segments each of length l_n. Then we have:

$$N_1 = 3 \text{ and } N_n = 4 \cdot N_{n-1}\ \forall\ n \ge 2.$$

Also,

$$l_1 = 1 \text{ and } l_n = \frac{l_{n-1}}{3}\ \forall\ n \ge 2.$$

The two recursions yield

$$N_n = 3 \cdot 4^{n-1} \text{ and } l_n = 3^{1-n}\ \forall\ n \ge 1.$$

Therefore the perimeter of K_n equals $N_n \cdot l_n = 3 \cdot \left(\frac{4}{3}\right)^{n-1}$ which diverges to ∞ as $n \to \infty$. □

Proof that K has zero area. We will now prove that K has zero area by showing that K_1' has zero area. We do this by covering K_1' with isosceles triangles as in [FU16].

Lemma 4.4.2. *Let K' be the Koch Curve generated by a line segment $\mathcal{L}((x_1,y_1),(x_2,y_2))$ of length a. Then:*

1. *The endpoints of K' are (x_1,y_1) and (x_2,y_2); hence the base length of K' is a.*
2. *For all $(x,y)\in K'$, the perpendicular distance from (x,y) to the base line $\mathcal{L}((x_1,y_1),(x_2,y_2))$ is at most*
$$\frac{1}{\sqrt{3}}\cdot\min\left\{\left\|(x,y)-(x_1,y_1)\right\|,\left\|(x,y)-(x_2,y_2)\right\|\right\}.$$

Proof of Lemma 4.4.2. We define a similarity map $T:\mathbb{R}^2\to\mathbb{R}^2$ such that

$$T(x_1,y_1)=(0,0),\quad T(x_2,y_2)=(1,0),$$

explicitly given by:
$$T(x,y):=\frac{1}{a}\begin{bmatrix}x_2-x_1 & y_2-y_1\\ -(y_2-y_1) & x_2-x_1\end{bmatrix}\begin{bmatrix}x-x_1\\ y-y_1\end{bmatrix}.$$

The similarity maps that act on the normalized segment $\mathcal{L}((0,0),(1,0))$ are:

$$S_1(x,y)=\left(\frac{x}{3},\frac{y}{3}\right),$$
$$S_2(x,y)=\left(\frac{1}{6}x+\frac{\sqrt{3}}{6}y+\frac{1}{3},\ -\frac{\sqrt{3}}{6}x+\frac{1}{6}y\right),$$
$$S_3(x,y)=\left(\frac{1}{6}x-\frac{\sqrt{3}}{6}y+\frac{1}{2},\ \frac{\sqrt{3}}{6}x+\frac{1}{6}y+\frac{\sqrt{3}}{6}\right),$$
$$S_4(x,y)=\left(\frac{x}{3}+\frac{2}{3},\frac{y}{3}\right).$$

The similarity transformations acting on the original base segment $\mathcal{L}((x_1,y_1),(x_2,y_2))$ are:
$$S'_j:=T^{-1}\circ S_j\circ T,\quad j=1,2,3,4.$$
We know that:
$$K'=\lim_{k\to\infty}\mathbf{S}'^k(\mathcal{L}((x_1,y_1),(x_2,y_2))).$$
Let $\Delta_0\subset\mathbb{R}^2$ denote the isosceles triangle with base $\mathcal{L}((0,0),(1,0))$ and height $\frac{\sqrt{3}}{6}$, that is,
$$\Delta_0:=\left\{(x,y)\in\mathbb{R}^2:\ 0\le x\le 1,\quad 0\le y\le\frac{\min(x,1-x)}{\sqrt{3}}\right\}.$$

We prove by induction that for all $k\ge 1$,
$$\mathbf{S}^k(L_1)\subset\Delta_0.$$

Base case: $k=1$. We have

- $S_1(L_1)=\mathcal{L}((0,0),(\frac{1}{3},0))\subset\Delta_0$ $\qquad S_2(L_1)=\mathcal{L}\left(\left(\frac{1}{3},0\right),\left(\frac{1}{2},\frac{\sqrt{3}}{6}\right)\right)\subset\Delta_0$
- $S_3(L_1)=\mathcal{L}\left(\left(\frac{1}{2},\frac{\sqrt{3}}{6}\right),\left(\frac{2}{3},0\right)\right)\subset\Delta_0$ $\qquad S_4(L_1)=\mathcal{L}\left(\left(\frac{2}{3},0\right),(1,0)\right)\subset\Delta_0$

Hence, $\mathbf{S}^1(L_1)\subset\Delta_0$.

Inductive step. Assume $\mathbf{S}^k(L_1) \subset \Delta_0$. Since each S_j is a contraction and satisfies $S_j(\Delta_0) \subset \Delta_0$, it follows:

$$\mathbf{S}^{k+1}(L_1) = \bigcup_{j=1}^{4} S_j\left(\mathbf{S}^k(L_1)\right) \subset \Delta_0.$$

Hence, by induction, $K_1' = \lim_{k\to\infty} \mathbf{S}^k(L_1) \subset \Delta_0$.

Since T is a linear map, we have

$$\mathbf{S'}^{k+1}(\mathcal{L}((x_1,y_1),(x_2,y_2)) = \bigcup_{j=1}^{4} S_j'\left(\mathbf{S'}^k(\mathcal{L}((x_1,y_1),(x_2,y_2))\right) \subset \Delta,$$

where Δ is the isosceles triangle with base $\mathcal{L}((x_1,y_1),(x_2,y_2))$.

Now,

$$K' := \lim_{k\to\infty} \mathbf{S'}^k(\mathcal{L}((x_1,y_1),(x_2,y_2))) \subset \bigcup_{k=1}^{\infty} \mathbf{S'}^k(\mathcal{L}((x_1,y_1),(x_2,y_2))) \subset \Delta$$

by definition.

Thus K' is completely contained inside an isosceles triangle of base length a and height $\frac{a}{2\sqrt{3}}$ and thus the area of $K' < \frac{1}{2}\cdot a \cdot \frac{a}{2\sqrt{3}} = \frac{a^2}{4\sqrt{3}}$. □

1. Iteration 1: K_1' is generated by a line of length 1 and thus area of $K_1' < \frac{1}{4\sqrt{3}}$.
2. Iteration 2: Now, we divide K_1' into 4 Koch curves, each generated by a line of length $\frac{1}{3}$. Using this approximation, we can say that the area of $K_1' < 4 \cdot \frac{\frac{1}{9}}{4\sqrt{3}} = (\frac{4}{9})^1 \cdot \frac{1}{4\sqrt{3}}. \dots$
3. Iteration $n+1$: Let the n^{th} iteration give us 4^{n-1} Koch curves, each generated by a line of length $\frac{1}{3^{n-1}}$ and hence the upper bound $(\frac{4}{9})^{n-1} \cdot \frac{1}{4\sqrt{3}}$. Now, we divide each of the Koch Curves into 4 Koch curves, each generated by a line of length $\frac{1}{3^n}$. Using this approximation, we can say that the area of $K_1' < 4 \cdot (\frac{4}{9})^{n-1} \cdot \frac{1}{9} \cdot \frac{1}{4\sqrt{3}} = (\frac{4}{9})^n \cdot \frac{1}{4\sqrt{3}}$.

Continuing this process tells us that that the area of $K_1' < (\frac{4}{9})^n \cdot \frac{1}{4\sqrt{3}}\ \forall\ n \in \mathbb{N}$. *As* $n \to \infty$, we can say that the area of $K_1' \leq 0$. Since area is always non-negative, we have that K_1' has area 0. We repeat this process for K_2' and K_3' and conclude that $K = K_1' \cup K_2' \cup K_3'$ has area 0. □

Thus K is a set with infinite perimeter and zero area, which shows that its dimension must lie somewhere between 1 and 2. In this section we will prove the result stated below. Let K be the Koch Snowflake. Then

Theorem 4.4.3. *(a)* $\dim_T(K) = 1$,

(b) $\dim_B(K) = \log_3(4)$ *and*

(c) $\dim_H(K) = \log_3(4)$.

4.4.3 Proof of Theorem 4.4.3(a)

We will show that K has a topological dimension of 1. In order to show this we will show first that $\dim_T(K) \geq 1$ and then show that $\dim_T(K) \leq 1$.

Proof that $\dim_T(K) \geq 1$. We will show that the Koch Snowflake is compact. Since $K \subset \mathbb{R}^2$, by Theorem 2.0.13 it is enough to show that K is closed and bounded.

Step 1 K **is bounded** Given the vertices of K_1 are $(0,0),(1,0)$ and $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, the circumcentre of K_1 is at $r_0 = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right)$. We will show that K is bounded by the $B\left(r_0, \frac{1}{\sqrt{3}}\right)$, where $\frac{1}{\sqrt{3}}$ is the circumradius of K_1. The length of the perpendicular bisector from the circumradius on one of the sides is $\frac{r}{2\sqrt{3}}$ where r is the side length of the triangle. Hence, at each iteration, the new point added is at a distance of

$$d \leq \frac{l_0}{2\sqrt{3}} + \frac{\left(\frac{l_0}{3}\right)}{2\sqrt{3}} + \frac{\left(\frac{l_0}{3^2}\right)}{2\sqrt{3}} + \dots$$

from 0 where $l_0 = 1$, the side length of K_1. Therefore, $d \leq \frac{3}{4\sqrt{3}} < \frac{1}{\sqrt{3}}$. Therefore the new points added are still contained in $B\left(r_0, \frac{1}{\sqrt{3}}\right) \implies K$ is bounded.

Step 2 K **is closed** Firstly, K_n is closed for finite n as K_n is a polygonal curve, formed by finitely many line segments. Each line segment is a closed subset of $\mathbb{R}^2$ and the union of finitely many closed sets is closed. Therefore, each K_n is closed. We will show:

$$K = \lim_{n\to\infty} K_n = \bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n}$$

(a) **Inclusion** $K \subseteq \bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n}$. Let $x \in K$. Then there exists a sequence $\{x_n\}$ with $x_n \in K_n$ and $x_n \to x$. For any fixed m, since $x_n \to x$, eventually $x_n \in \bigcup_{n\geq m} K_n$. Hence, $x \in \overline{\bigcup_{n\geq m} K_n}$ for every m, so:

$$x \in \bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n}$$

(b) **Inclusion** $\bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n} \subseteq K$. Let $x \in \bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n}$. Then for every m, there exists a point $x_m \in \bigcup_{n\geq m} K_n$ such that:

$$\|x - x_m\| < \frac{1}{m}$$

Thus, $x_m \to x$ and each $x_m \in K_{n_m}$ for some $n_m \geq m$. Hence, x is the limit of a sequence $x_m \in K_{n_m}$ and so $x \in K$ by definition. Therefore:

$$\bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n} \subseteq K$$

Combining the two inclusions, we conclude:

$$K = \bigcap_{m=0}^{\infty} \overline{\bigcup_{n\geq m} K_n}$$

Hence K is the arbitrary intersection of closures of sets, which are closed $\implies$ K is closed.

Now we shall show that K is **connected**. We will show that the Koch Curve K' is connected. Since the Koch Snowflake is the union of three Koch Curves attached at the boundaries, by applying Lemma 2.0.20, it is enough to prove that the Koch Curve is connected in order to prove that the Koch Snowflake is connected. We assume for the sake of contradiction that K' is disconnected. Then there exist relatively open subsets $A, B \subset K'$ such that:

1. $K' = A \cup B$.
2. $A \cap B = \emptyset$.
3. $A \neq \emptyset$ and $B \neq \emptyset$.

By definition of relative openness, there exist open sets $U, V \subset \mathbb{R}^2$ such that:

$$A = K' \cap U \quad \text{and} \quad B = K' \cap V.$$

Since K' is compact (by Theorem 2.0.13) and A, B are relatively closed in K' (since their complements B and A are relatively open in K'), A and B are compact in K'. By Lemma 2.0.14:

$$\delta := d(A,B) = \inf\{\|a-b\| : a \in A, b \in B\} > 0.$$

Let K'_n be the n^{th} approximation of K'. Then K'_n has 4^n line segments of length 3^{-n+1} each. Let the line segments be denoted by $\{l_1, l_2, \ldots, l_{4^n}\}$. We choose n large enough so that the lengths of the line segments satisfies $3^{-n+1} < \delta/2$.

$\forall\ i \in [4^n]$, for line segment $l_i \subset K'_n$ intersecting K':

- $\operatorname{diam}(l_i \cap K') = 3^{-n+1} < \delta/2$.
- If $l_i \cap K' \cap A \neq \emptyset$ and $l_i \cap K' \cap B \neq \emptyset$, then $\exists\ a \in A,\ b \in B$ with $d(a,b) \leq \operatorname{diam}(l_i) < \delta$, contradicting $d(A,B) = \delta$.

Thus each $l_i \cap K'$ lies entirely in A or B.

This would require K'_n to be disconnected if both A and B are non-empty. However, K'_n is the union of 4^n connected line segments $l_i's$ where l_i and l_{i+1} have their endpoints in intersection for all $i \in [4^n - 1]$. Hence, by Lemma 2.0.20, K'_n is connected and we arrive at a contradiction.

Since it was enough to prove the connectedness of K', we have proved that K is also connected which along with compactness shows that the topological dimension of $K \geq 1$ by Lemma 2.0.28. □

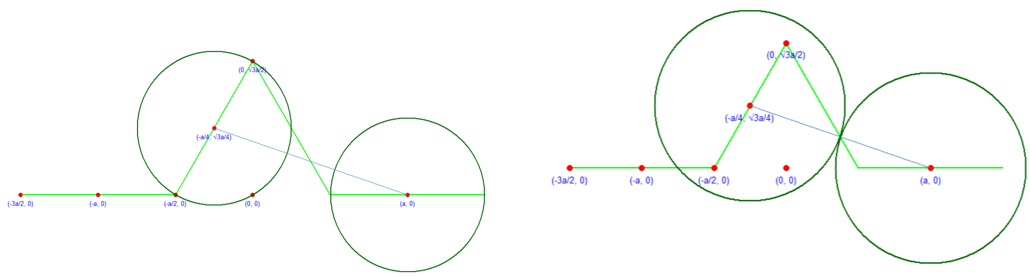
Proof that $\dim_T(K) \leq 1$. Due to compactness of K, given an arbitrary open cover $\mathcal{U}$, it has a positive Lebesgue number δ such that any set $V \subset K$ with $\operatorname{diam}(V) < \delta$ is completely contained inside some element of $\mathcal{U}$.

We choose $n \ni d_n := 3^{-n+1} < \frac{2\delta}{\sqrt{7}}$ and consider K_n, the n^{th} polygonal approximation of K.

K_n has $N_n := 3 \cdot 4^{n-1}$ line segments of length 3^{-n+1} each. Let $\{x_1, x_2, \ldots, x_{N_n}\}$ be the midpoints of the line segments $\{l_1, l_2, \ldots, l_{N_n}\}$.

Say for some $r > 0$, we construct open balls $B_i := B(x_i, r) \forall\ 1 \leq i \leq N_n$. Then the minimum distance between B_i and B_{i+2} will be given by $d - 2r$ where d is the distance between their centres. Now, the distance between x_i and x_{i+2} can be calculated to be $d = \frac{\sqrt{7} \cdot 3^{-n+1}}{2}$. If we have three sets in $\{B_i\}_{i=1}^{N_n}$ with non-empty intersection, our refinement will be of order greater than 2. Therefore we must have $d - 2r \geq 0 \implies r \leq \frac{\sqrt{7} \cdot 3^{-n+1}}{4}$ as shown in Figure 4.4b. Also, in order to cover the entire l_i with x_i, we must have $r > \frac{3^{-n+1}}{2}$ as in Figure 4.4a. Let r_0 satisfy

$$\frac{3^{-n+1}}{2} < r_0 \leq \frac{\sqrt{7} \cdot 3^{-n+1}}{4}.$$



(a) B_i and B_{i+2} when $r_0 = \frac{3^{-n}}{2}$

(b) B_i and B_{i+2} when $r_0 = \frac{\sqrt{7} \cdot 3^{-n}}{4}$

Figure 4.4: B_i and B_{i+2} where $a = 3^{-n}$

The open balls $B_i := B(x_i, r_0)$ for $1 \leq i \leq N_n$ thus cover B_i such that $B_i \cap B_{i+2} = \emptyset$ $\forall\ 1 \leq i < N_n - 1$ and $B_{N_n - 1} \cap B_1 = B_{N_n} \cap B_2 = \emptyset$.

We have also shown that all sets in $\{B_i\}_{i=1}^{N_n}$ have diameter lesser than δ as $\forall\ 1 \leq i \leq N_n$, $\mathrm{diam}(B_i) = 2r_0 < \frac{\sqrt{7} \cdot 3^{-n}}{2} < \delta$. By the definition of Lebesgue number, all $B_i's$ are contained in some $\mathcal{U}_j \in \mathcal{U}$. Then $\mathcal{V} := \{B_i \cap K_n\}_{i=1}^{N_n}$ forms an open cover of K_n as all line segments are covered by some B_i.

In the following iterations, the new points added to K_n from l_i are at a distance of

$$d < \frac{\sqrt{3} d_n}{2 \cdot 3} + \frac{\sqrt{3} d_n}{2.3^2} + \frac{\sqrt{3} d_n}{2.3^3} + \ldots < \frac{\sqrt{3} d_n}{4} < \frac{3^{-n+1}}{2} < r_0$$

from x_i. Hence, the new points added in the iterations following K_n all lie inside some B_i.

Therefore, the set $\mathcal{V}$ forms an open cover of K. Also, the distance between any two non-consecutive $B_i's$ is positive by choice of r_0, meaning each point of K lies in at most two sets of $\mathcal{V}$, meaning $\mathcal{V}$ has order 2. Therefore, the topological dimension of $K \leq 1$. $\square$

We have thus shown that $\dim_T(K) = 1$. $\square$

Below, we will calculate the two fractal dimensions, namely the box and Hausdorff dimensions and show that they are all equal to $\log_3(4) \in (1, 2)$.

4.4.4 Proof of Theorem 4.4.3(b)

By Definition 2.0.32, the box dimension of K is given by

$$-\lim_{\epsilon\to 0}\frac{\log(N(\epsilon))}{\log(\epsilon)}$$

where $N(\epsilon)$ is the minimum number of boxes of size $\epsilon > 0$ needed to cover K. We will estimate $N_n(\epsilon)$, the minimum number of boxes of size ϵ required to cover K_n, the n^{th} polygonal approximation of K. Since we are concerned about the limit as $\epsilon \to 0$, we can assume $\epsilon \leq 3^{-n}$, given n. This ensures that the size of the box is lesser than each individual line segment. By a method similar to Lemma 3.5, it can be shown that boxes of side ϵ placed such that the line segments l_i pass through the diagonals minimizes $N_n(\epsilon)$. K_n has $3\cdot 4^{n-1}$ line segments of length 3^{-n+1} each. Thus, it has a perimeter of $3\cdot(\frac{4}{3})^{n-1}$.

Then $\exists\ t \geq 1$ such that $N_n(\epsilon) = t\cdot\frac{3\cdot(\frac{4}{3})^{n-1}}{\sqrt{2}\epsilon}$, where $(3\cdot\frac{4}{3})^{n-1}$ is the perimeter of K_n and $\sqrt{2}\epsilon$ is the diagonal of one box. We now compute

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = -\frac{(n-1)\log\left(\frac{4}{3}\right)+\log(3t)-\log(\sqrt{2})-\log(\epsilon)}{\log(\epsilon)}.$$

We can assume $\epsilon = c\cdot 3^{-n}$ for some $0 < c \leq 1$. Then

$$-\frac{\log(N_n(\epsilon))}{\log(\epsilon)} = -\frac{(n-1)\log\left(\frac{4}{3}\right)+\log(3t)-\log(c\sqrt{2})-\log(3^{-n})}{\log(c\cdot 3^{-n})}.$$

By simple computation, it is easy to see that as $n\to\infty$, the limit above exists and is equal to $\log_3(4)$. Hence, $\dim_B(K) = \log_3(4) \approx 1.2618\ldots$.

□

4.4.5 Proof of Theorem 4.4.3(c)

By Definition 2.0.34, we need $d \ni H^s(K) = 0\ \forall\ s > d$ and $d \ni H^s(K) = \infty\ \forall\ s < d$. We claim that $\dim_H(K) = \log_3(4)$. In order to show that we will show first that $\dim_H(K) \geq \log_3(4)$ and subsequently show that $\dim_H(K) \leq \log_3(4)$.

Proof of $\dim_H(K) \geq \log_3(4)$. We use Frostman's Lemma for this part of the proof. We define $\mu_n : \mathcal{B}(K_n)$ as a uniform mass distribution over K_n such that

$$\mu(l_i) = \frac{1}{3\cdot 4^{n-1}}\ \forall\ i,$$

where l_i is the i^{th} line segment of K_n. Then μ_n can be extended to $\mu : \mathcal{B}(K) \to [0,1]$ by

$$\mu(E) = \lim_{n\to\infty}\mu_n(E\cap K_n)\ \forall\ E \in \mathcal{B}(K).$$

Then given $r < 1$, we can choose n such that

$$3^{-n+1} \leq r < 3^{-n}.$$

The length of each line segment l_i of K_n is 3^{-n+1}. Thus the diameter of $B(x,r)$ is lesser than $\left(\frac{2}{3}\right)^{rd}$ the length of one line segment of K_n. Then observe that there exists a finite $c > 0$ such that $B(x,r)$ contains parts of c line segments of K_n for all $x \in K_n$ and for all such r. Then

$$\mu(B(x,r)) \ \leq \ \frac{c}{3 \cdot 4^{n-1}} \ = \ \frac{c}{3} \cdot \left(\frac{1}{3^{n-1}}\right)^{\log_3(4)} \leq \frac{c}{3} r^{\log_3(4)}.$$

And when $r > 1$, we have

$$\mu(B,r) \ \leq \ 1 \ \leq \ r^{\log_3(4)}.$$

Therefore by the Frostman's lemma we can say that $\dim_H(K) \ \geq \ \log_3(4)$. $\square$

Proof of $\dim_H(K) \leq \log_3(4)$. It is enough to show that for all $s > \log_3(4)$, $H^s(K) = 0$. Say $s = \log_3(4) + k$ for some $k > 0$.

Given $\delta > 0$, let $\mathcal{P}_\delta$ be the set of all countable open covers of K, $\mathcal{U} = \{\mathcal{U}_i\}_{i\in I}$ satisfying $\operatorname{diam}(\mathcal{U}_i) \leq \delta \ \forall \ i \in I$, with I being a suitable index set.

By Definition 2.0.34,

$$H^s(K) = \lim_{\delta\downarrow 0} \inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_{i\in I} (\operatorname{diam}(\mathcal{U}_i))^s$$

where $\sum_{i\in I}(\operatorname{diam}(\mathcal{U}_i))^s$ is a finite sum if I is finite and $\sum_{i\in I}(\operatorname{diam}(\mathcal{U}_i))^s$ is defined to be $\lim_{n\to\infty} \sum_{\substack{i\in I_n \\ |I_n|<\infty \\ I_n\uparrow I}} (\operatorname{diam}(\mathcal{U}_i))^s$ if I is countably infinite. Let $\delta > 0$ be fixed. Then

$$\inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_{i} (\operatorname{diam}(\mathcal{U}_i))^s \ \leq \ \sum_{i=1}^{N(\delta)} \delta^s$$

where $N(\delta)$ is the minimum number of open sets of diameter δ required to cover K. In order to calculate $N(\delta)$, we need the minimum number of open sets of diameter δ, $N_n(\delta)$, that can cover K_n. Comparing the perimeter of K_n and the diameter of a ball of radius $\frac{\delta}{2}$,

$$N_n(\delta) = t \cdot \frac{3 \cdot \left(\frac{4}{3}\right)^{n-1}}{\left(\frac{\pi\delta^2}{4}\right)} \text{ for some } t > 0.$$

Let $n \in \mathbb{N}$ be fixed. Since we are concerned with $\delta \ \to \ 0$, we choose $\delta > 0$ such that

$$\delta = c \cdot 3^{-n} \text{ for some } c \in (0,1].$$

Then

$$H^s(K) := \lim_{\delta\to 0} \inf_{\mathcal{U}\in\mathcal{P}_\delta} \sum_{i\in I} (\operatorname{diam}(\mathcal{U}_i))^s \leq \lim_{\substack{n\to\infty \\ \delta\to 0}} 3t \cdot \left(\frac{4}{3}\right)^{n-1} \cdot \delta^{s-1} \ni \delta = c \cdot 3^{-n}.$$

It is easy to see that the limit above tends to 0 which shows that $H^s(K) \ = \ 0$. $\square$

Thus we have shown that $\dim_H(K) = \log_3(4)$. $\square$

We have hence proved Theorem 4.4.3. $\square$

Chapter 5

The Similarity Dimension

In the previous sections we have dealt with three notions of dimensions, namely the topological, box and Hausdorff dimensions. In this section we will discuss another dimension, namely the Similarity dimension. This dimension, although easy to compute is not dealt with as much. You may refer to [FU16] and [Bre14] for discussions. We will define some concepts required to understand the similarity dimension first.

Definition 5.0.1. (Linear Map) *Let $X \subset \mathbb{R}^m$. A map $L : X \mapsto \mathbb{R}^n$ is said to be **linear** if*

$$L(\alpha x_1 + \beta x_2) = \alpha L(x_1) + \beta L(x_2)$$

for all $\alpha, \beta \in \mathbb{R}$ and $x_1, x_2 \in X$ such that $\alpha x_1 + \beta x_2 \in X$.

Definition 5.0.2. (Affine Map) *Let $X \subset \mathbb{R}^m$. A map $f : X \mapsto \mathbb{R}^n$ is said to be **affine** if there exists a linear map $L : X \mapsto \mathbb{R}^n$ and a translation vector $u \in R^n$ such that for all $x \in X$,*

$$f(x) = L(x) + u.$$

Definition 5.0.3. (Contraction Map) *Let $X \subset \mathbb{R}^m$. A map $g : X \mapsto \mathbb{R}^n$ is said to be a c−**contraction** for some $c \in [0, 1)$ if for all $x_1, x_2 \in X$,*

$$\|(g(x_1) - g(x_2)\| \leq c\|x_1 - x_2\|.$$

Definition 5.0.4. (Similarity Map) *Let $X \subset \mathbb{R}^m$. A map $g : X \mapsto \mathbb{R}^n$ is said to be a c−**similarity** for some $c \geq 0$ if for all $x_1, x_2 \in X$,*

$$\|(g(x_1) - g(x_2)\| = c\|x_1 - x_2\|.$$

Definition 5.0.5. (Self-Similar Set) *Let $X \subset \mathbb{R}^n$ be compact. We call a compact set $F \subset X$ a **self-similar** subset of X if for some $c \in [0, 1)$ there exists a set of c−**similarity affine contractions** $\mathcal{S} = \{S_i : X \to \mathbb{R}^n \mid i \in I\}$ such that*

$$F = \bigcup_{i \in I} S_i(F).$$

*F is then called the **attractor** of the set of maps $\mathcal{S}$.*

We use a lemma in order to define the similarity dimension of a self-similar set.

Lemma 5.0.6. *Let $X \subset \mathbb{R}^n$ be compact and $\mathcal{S} = \{S_i : X \to \mathbb{R}^n \mid i \in I\}$ be a set of c–similarity affine contractions for some $c \in [0,1)$.*

(a) Then there exists a unique attractor $F \subset X$.

(b) Let $Y \subset X$ be compact such that $S_i(Y) \subset Y \ \forall \ i \in I$. Let us define a sequence of sets $\{Y_n\}_{n\in\mathbb{N}} \subset \mathcal{P}(Y)$ as follows

- $\to$ $Y_0 = Y$ *and*
- $\to$ $Y_n = \bigcup_{i\in I} S_i(Y_{n-1}) \ \forall \ n \geq 1$.

where $\mathcal{P}(Y)$ is the power set of Y. We can then obtain the unique attractor F of the set of maps $\mathcal{S}$ by

$$F = \bigcap_{n=0}^{\infty} Y_n.$$

Refer to Theorem 9.1 of [Fal13] for proof.

Definition 5.0.7. (Self-Similar Copy) *Let $X, \mathcal{S}, Y, \{Y_n\}_{n\in\mathbb{N}}$ and F be as in Lemma 5.0.6. For some $n \in \mathbb{N}$, a compact set $Z \subset X$ is said to be a **self-similar copy** of Y_n if there exists $i_1, i_2, \ldots, i_k \in I$ such that*

$$Z = S_{i_k} \circ S_{i_{k-1}} \circ \ldots \circ S_{i_1}(Y_n).$$

We are now equipped with the machinery to define the similarity dimension of a self-similar set.

Definition 5.0.8. (Similarity Dimension) *Let $X, \mathcal{S}, Y, \{Y_n\}_{n\in\mathbb{N}}$ and F be as in Lemma 5.0.6. For given $k \in \mathbb{N}$, we define N_k to be the number of self-similar copies of Y_0 that are subsets of Y_k. Since $Y_k := \bigcup_{i\in I} S_i(Y_{k-1})$ for all $k \geq 1$, we have*

$$N_k = \mid I \mid \cdot N_{k-1} \ \forall \ k \geq 1.$$

For given $k \in \mathbb{N}$, we define l_k to be the diameter of each self-similar copy of Y_0 that is a subset of Y_k. Since $Y_k := \bigcup_{i\in I} S_i(Y_{k-1})$ for all $k \geq 1$ and S_i's are all c–similarities for some $c \in [0,1)$, we have

$$l_k = c \cdot l_{k-1} \ \forall \ k \geq 1.$$

*The **similarity dimension** of F is given by*

$$\dim_S(F) = \log_{\frac{1}{c}} \mid I \mid.$$

We will now compute the similarity dimension for the sets discussed above, namely the interval I, the box B, the Cantor Set C^α, the Sierpiński Gasket S_3, the Sierpiśki Carpet S_4 and the Koch Snowflake K.

Interval I: We define $I_0 = I$ and maps $S_1, S_2 : I \to I$ as

$$S_1(x) = \frac{x}{2} \text{ and } S_2(x) = \frac{x}{2} + \frac{1}{2}$$

for all $x \in I$. It is easy to see that $I_n := S_1(I_{n-1}) \cup S_2(I_{n-1}) = I \ \forall \ n \geq 0$. Thus $I = \bigcap_{n=0}^{\infty} I_n$.

At the n^{th} iteration, we have $N_n = 2N_{n-1}$ line segments of length $l_n = \frac{l_{n-1}}{2}$. Thus the similarity dimension of I is $\frac{\log 2}{\log 2} = 1$. □

Box B**:** We define $B_0 = B$ and maps $\{S_i\}_{i=1}^4 : B \to B$ as

$$S_1(x,y) = \left(\frac{x}{2}, \frac{y}{2}\right), \quad S_2(x,y) = \left(\frac{x}{2} + \frac{1}{2}, \frac{y}{2}\right),$$
$$S_3(x,y) = \left(\frac{x}{2}, \frac{y}{2} + \frac{1}{2}\right), \quad S_4(x,y) = \left(\frac{x}{2} + \frac{1}{2}, \frac{y}{2} + \frac{1}{2}\right)$$

for all $(x,y) \in B$. It is easy to see that $B_n := \bigcup_{i=1}^4 S_i(B_{n-1}) = B \; \forall \; n \geq 0$. Thus $B = \bigcap_{n=0}^\infty B_n$.

At the n^{th} iteration, we have $N_n = 4N_{n-1}$ boxes of side length $l_n = \frac{l_{n-1}}{2}$. Thus the similarity dimension of B is $\frac{\log 4}{\log 2} = 2$. □

Note that the circle S^1 is not self-similar in nature and thus similarity dimension is not defined for it.

$\alpha-$**Cantor Set** C^α**:** As we have seen in Definition 4.1.1, for some $\alpha \in (0,1)$, the $\alpha-$Cantor Set is constructed using **two** $\frac{1-\alpha}{2}-$similarity maps. By definition, the similarity dimension of C^α is $\log_{\frac{2}{1-\alpha}}(2)$. □

Sierpiński Gasket S_3**:** As we have seen in Definition 4.2.1, the Sierpiński Gasket is constructed using **three** $\frac{1}{2}-$similarity maps. By definition, the similarity dimension of S_3 is $\log_2(3)$. □

Sierpiński Carpet S_4**:** As we have seen in Definition 4.3.1, the Sierpiński Carpet is constructed using **eight** $\frac{1}{3}-$similarity maps. By definition, the similarity dimension of S_4 is $\log_3(8)$. □

Koch Snowflake K**:** As we have seen in Definition 4.4.1, the Koch Snowflake is constructed using **four** $\frac{1}{3}-$similarity maps. By definition, the similarity dimension of K is $\log_3(4)$. □

Hence we are done calculating the similarity dimensions of the sets discussed and with this we come to the end of our report.

Chapter 6

Conclusion

Now that we are familiar with various notions of dimension, we can define fractals precisely. We use the definition posed by Mandelbrot in [Can84].

1. A **fractal** is by definition a set for which the Hausdorff dimension strictly exceeds the topological dimension.

2. Every set with a non-integer Hausdorff Dimension is a fractal.

However, a fractal may have an integral Hausdorff Dimension, as in the trail of Brownian Motion which has a topological dimension of 1 and a Hausdorff dimension of 2 (see Chapter 25 of [Can84]). This study has demonstrated the diverse landscape of dimension theory as it applies to both classical and fractal sets. Through detailed analysis of examples ranging from the interval $[0,1]$ to complex self-similar structures such as the Koch snowflake and Sierpiński carpet, we have illustrated how topological, box-counting, similarity and Hausdorff dimensions reveal different aspects of a set's geometry and scaling behavior.

In particular, the occurrence of non-integer dimensions in box-counting, similarity and Hausdorff frameworks underscores the inadequacy of classical integer-valued dimensions for describing fractals. The fact that many of these fractals exhibit infinite perimeter and zero area while retaining non-trivial dimensionality further motivates the use of generalized dimension theories. Hausdorff measures and tools like Frostman's lemma give a strong mathematical foundation for understanding these results.

Our findings show that dimension is not a single, simple idea—it depends on both the shape of the set and how it behaves from a measure-theoretic point of view. This way of thinking is important in many areas, such as dynamical systems, signal processing, image compression, and modeling natural patterns, where fractal structures often appear.

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